

Next-Generation IEEE 802.11 standards

Basics of the IEEE 802.11 MAC sub-layer

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IEEE 802.11 MAC sub-layer

Contents

- Carrier Sense Multiple Access basics
- IEEE 802.11 architecture
- Physical and virtual channel sensing
- Access methods in IEEE 802.11 MAC sub-layer
- IEEE 802.11 MAC basics



IEEE 802.11 MAC sub-layer

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Multiple access protocols

- single shared broadcast channel
- two or more simultaneous transmissions by nodes: interference
 - *collision* if node receives two or more signals at the same time

multiple access protocol

- distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit
- communication about channel sharing must use channel itself!
 - no out-of-band channel for coordination



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Carrier Sense Multiple Access (CSMA)

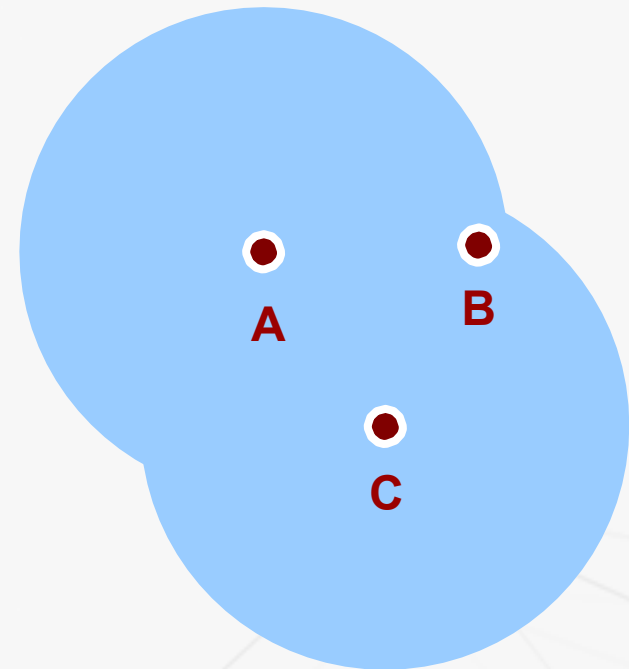
Simple CSMA: listen before transmit:

- if channel sensed idle: transmit entire frame
- if channel sensed busy: defer transmission

□ human analogy: don't interrupt others!

Carrier sensing (more detailed)

- A prospective sender listens to ("senses") the channel ("carrier") before trying to transmit.
- If the channel is idle, the sender is allowed to transmit.
- If the channel is busy (i.e. used by another device), the sender defers transmission for a certain time period ("backoff").
- The backoff period is determined by one of several strategies.



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CSMA backoff strategies

Non-persistent CSMA

- Sender draws a random waiting (backoff) time from a certain interval $[t_1, t_2]$
- After waiting, sender senses channel again

Persistent CSMA

- Sender continues to sense the channel and awaits end of current transmission
- Sender then either sends or behaves according to a backoff strategy

Example: *p*-persistent CSMA

- Once the channel is free, a prospective sender transmits with probability p and waits another fixed backoff period T with probability $(1-p)$
- Special case: 1-persistent



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CSMA and collisions



CSMA/CD: CSMA with *collision detection*

- collisions *detected* within short time
 - colliding transmissions aborted, reducing channel wastage
 - collision detection easy in wired, difficult with wireless
- human analogy: the polite conversationalist

- collisions can *still* occur with carrier sensing:
 - **propagation delay** means two nodes may not hear each other's just-started transmission
- **collision**: entire packet transmission time wasted
 - distance & propagation delay play role in determining collision probability

Source: Jim Kurose, Keith Ross, "Computer Networking: A Top-Down Approach", 9th edition Pearson, 2025.



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Multiple access links and protocols

Two types of “links”:

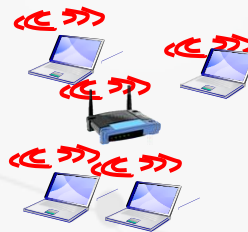
- point-to-point
 - point-to-point link between Ethernet switch, host
 - PPP for dial-up access
- **broadcast (shared wire or medium)**
 - old-school Ethernet
 - upstream HFC in cable-based access network
 - 802.11 wireless LAN, 4G/4G. satellite



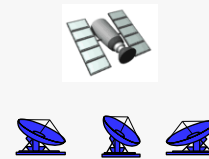
shared wire (e.g.,
cabled Ethernet)



shared radio: 4G/5G



shared radio: WiFi



shared radio: satellite



humans at a cocktail party
(shared air, acoustical)

Source: Jim Kurose, Keith Ross, “Computer Networking:
A Top-Down Approach”, 9th edition Pearson, 2025.



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Classification of MAC protocols

Centralized

- Dedicated network nodes (e.g. base stations) allocate channels

Reservation / Schedule-based

- Channel is reserved for dedicated use of a device pair
- Collisions do not occur

Distributed

- All network nodes “cooperate” to regulate channel access

Random Access / Contention-based

- Nodes compete for access to a shared channel in a probabilistic manner
- Collisions may occur and are resolved

Hybrid schemes



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Random access protocols

When node has packet to send:

- transmit at full channel data rate R
- no *a priori* coordination among nodes
- two or more transmitting nodes: “collision”

random access protocol specifies:

- how to detect collisions
- how to recover from collisions (e.g., via delayed retransmissions)

examples of random access MAC protocols:

- ALOHA, slotted ALOHA
- CSMA, CSMA/CD, CSMA/CA



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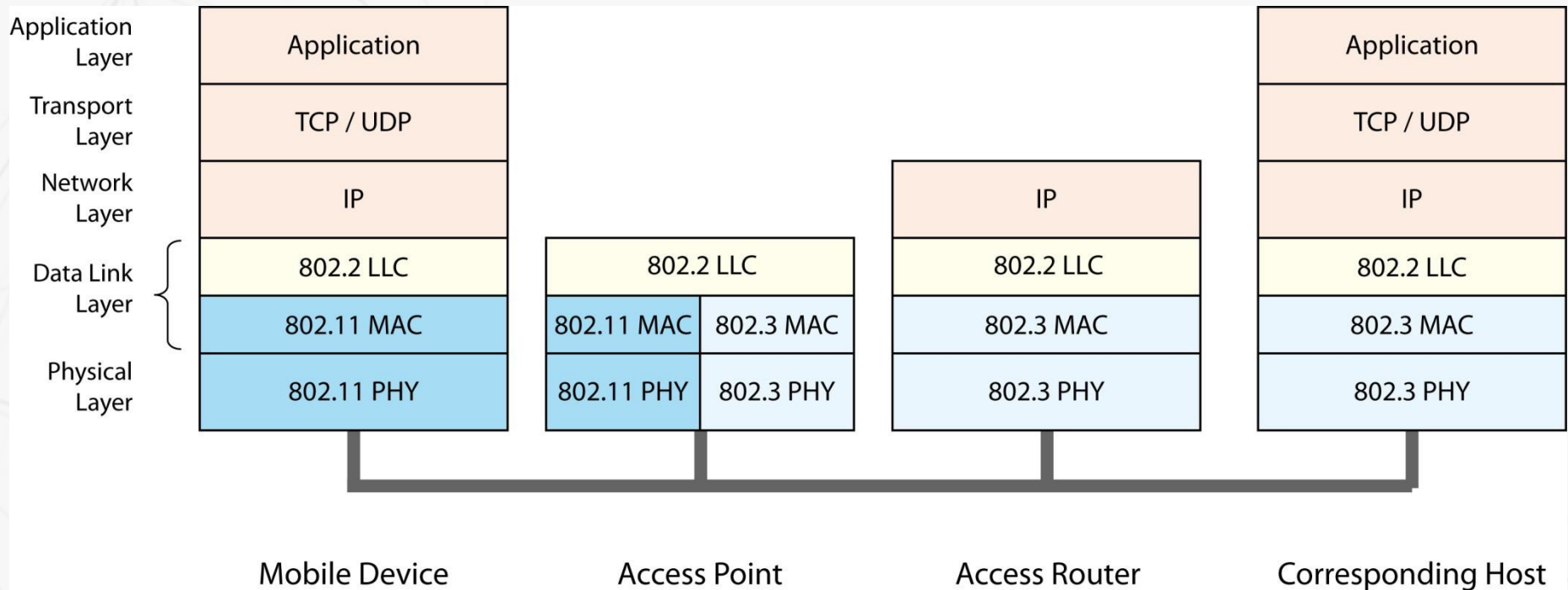
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IEEE 802.11 MAC sub-layer

Protocol stack of IEEE 802.11



Abbreviations:

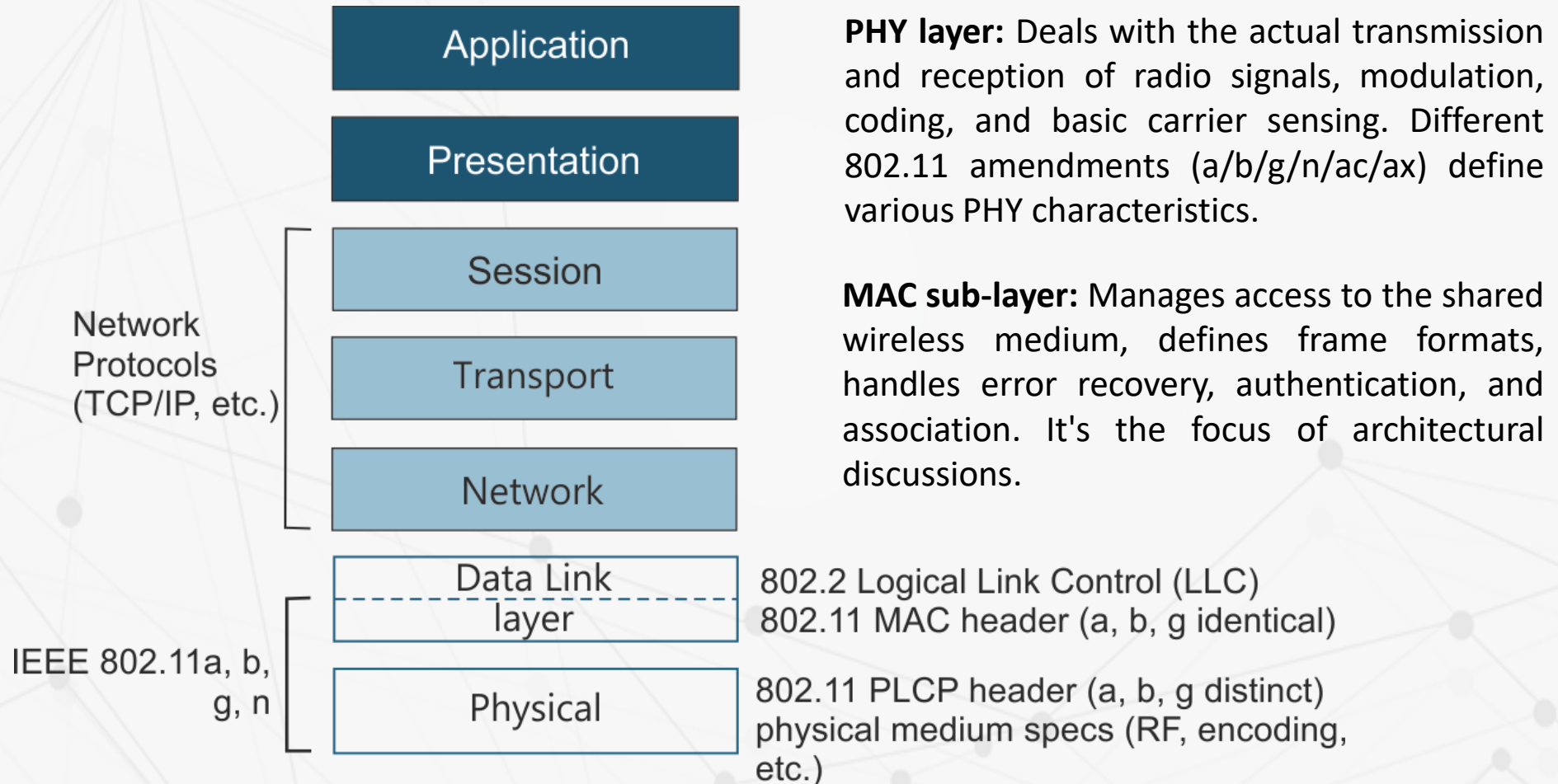
PHY Physical Layer
MAC Medium Access Control
LLC Logical Link Control

IP Internet Protocol
TCP Transfer Control Protocol
UDP User Datagram Protocol



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Protocol stack of IEEE 802.11



PHY layer: Deals with the actual transmission and reception of radio signals, modulation, coding, and basic carrier sensing. Different 802.11 amendments (a/b/g/n/ac/ax) define various PHY characteristics.

MAC sub-layer: Manages access to the shared wireless medium, defines frame formats, handles error recovery, authentication, and association. It's the focus of architectural discussions.



IEEE 802.11 MAC sub-layer

IEEE 802.11 MAC sub-layer overview

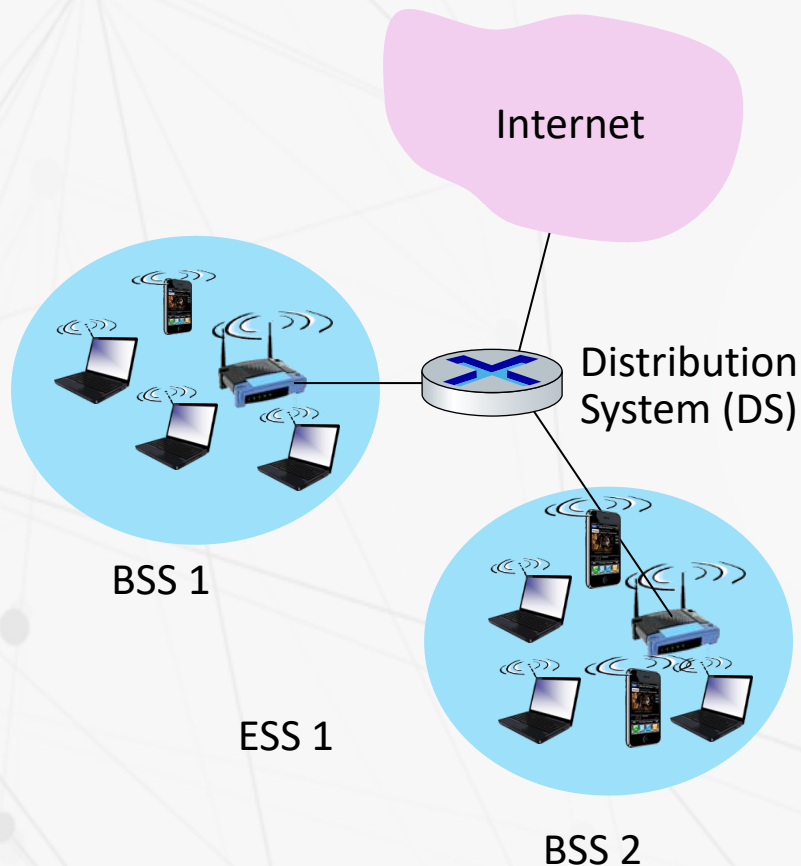
Key responsibilities of the IEEE 802.11 MAC:

- Medium access control: Determining who gets to transmit and when.
- Reliable data delivery: Providing mechanisms for acknowledgments, retransmissions, and fragment assembly/disassembly.
- Frame formatting: Defining the structure of IEEE 802.11 MAC frames (management, control, data).
- Association and authentication: Managing device connections and security.
- Power management: Enabling devices to conserve battery life.
- Quality of Service (QoS): Differentiating and prioritizing traffic types.



IEEE 802.11 MAC sub-layer

IEEE 802.11 LAN architecture



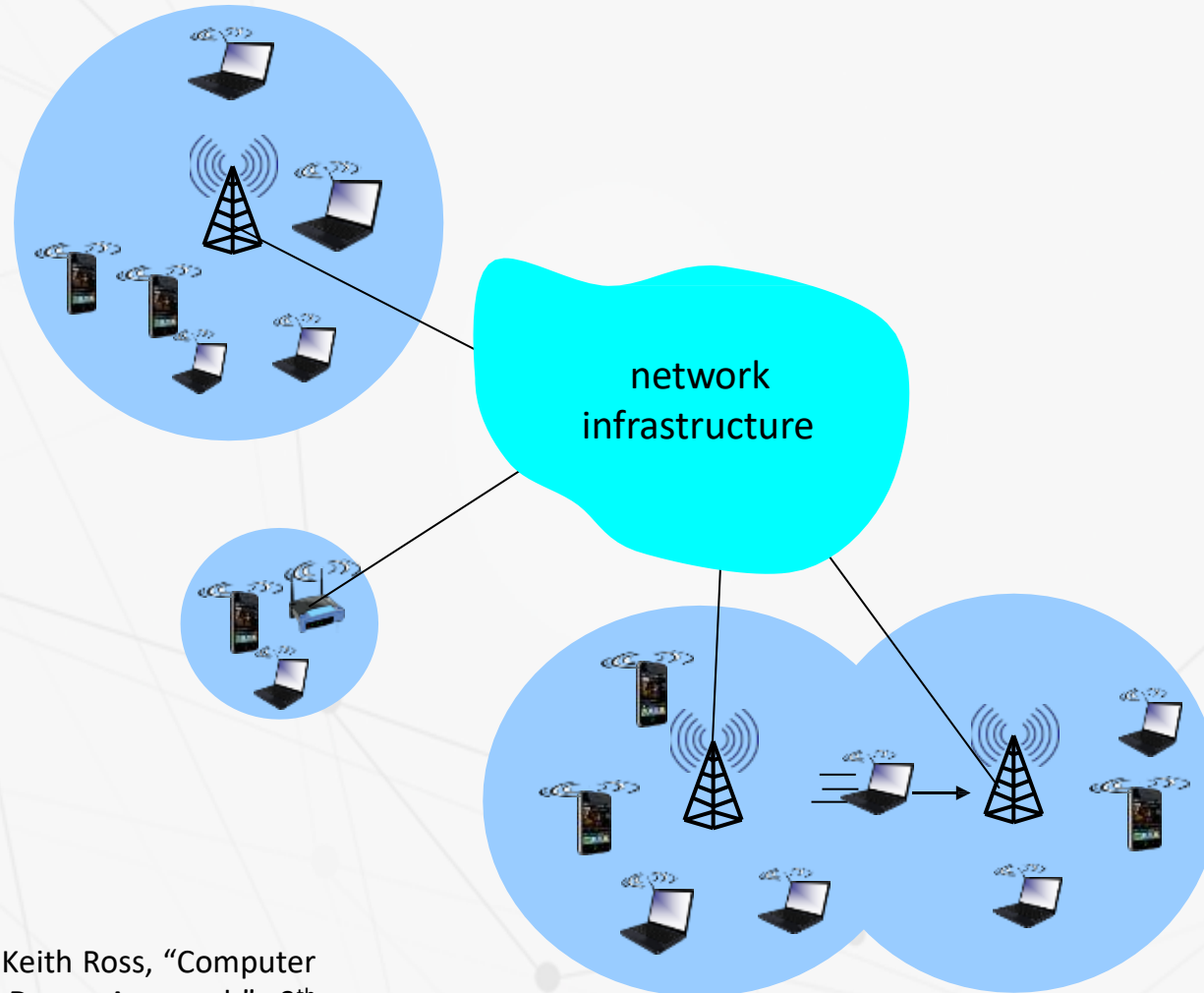
- wireless host communicates with base station
 - base station = access point (AP)
- Basic Service Set (BSS), the fundamental building block of an 802.11 network:
 - in infrastructure mode it contains wireless hosts and an Access Point (AP) -> base station
 - in ad hoc mode it contains hosts only
- Extended Service Set (ESS):
 - A set of connected BSSs forming a larger network

Source: Jim Kurose, Keith Ross, "Computer Networking: A Top-Down Approach", 9th edition Pearson, 2025.



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Elements of a wireless (local area) network

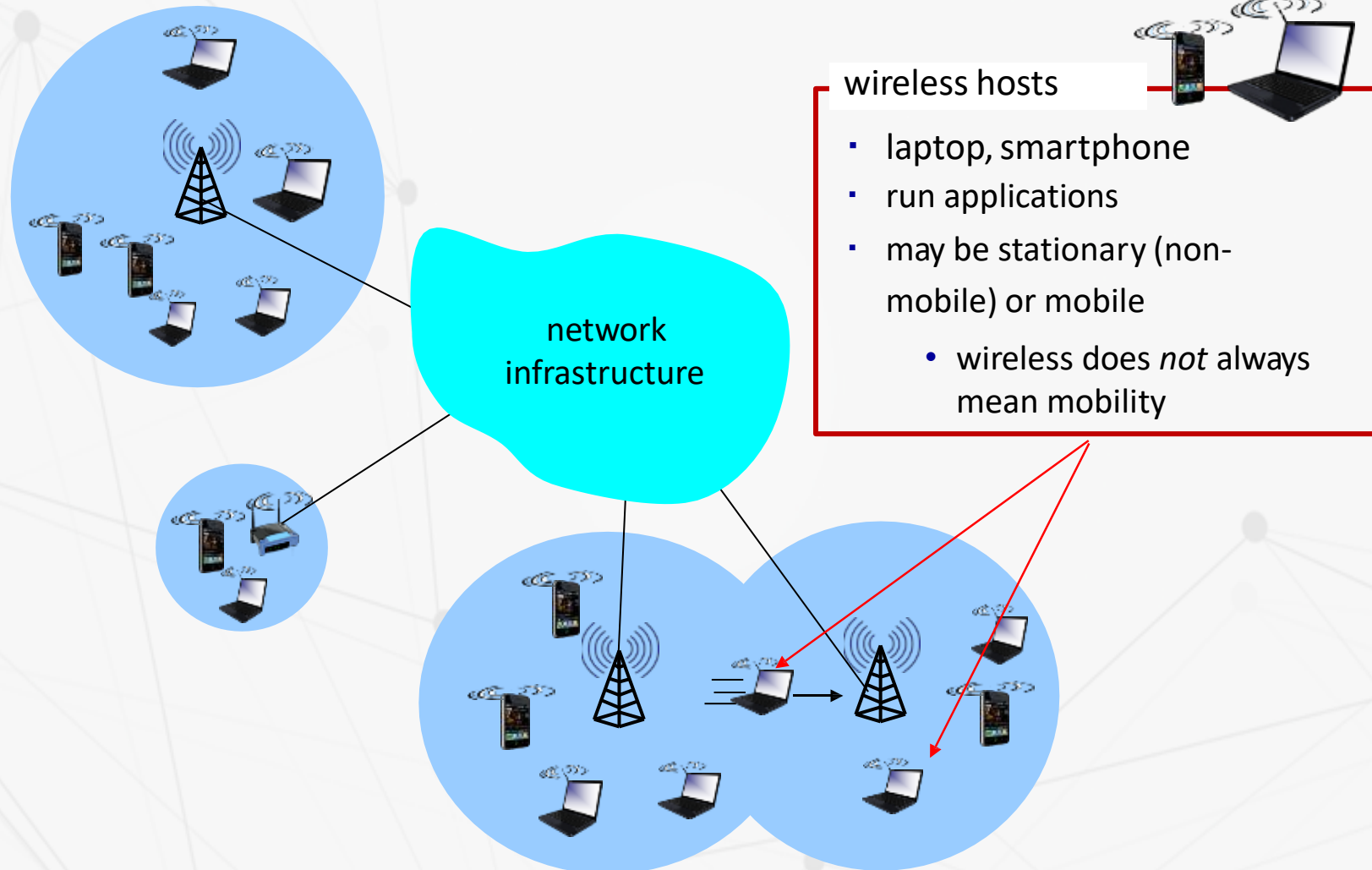


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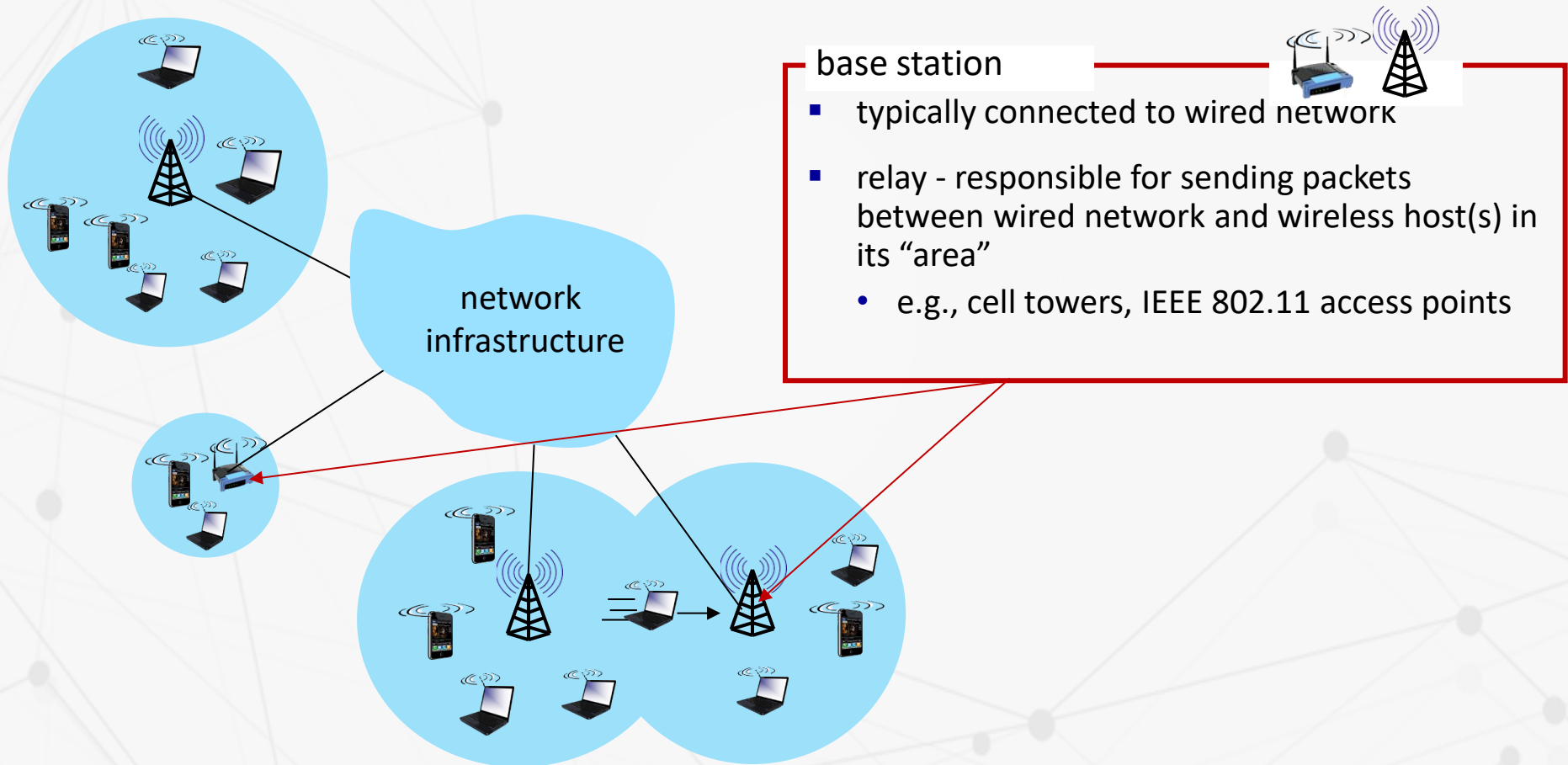
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Elements of a wireless (local area) network



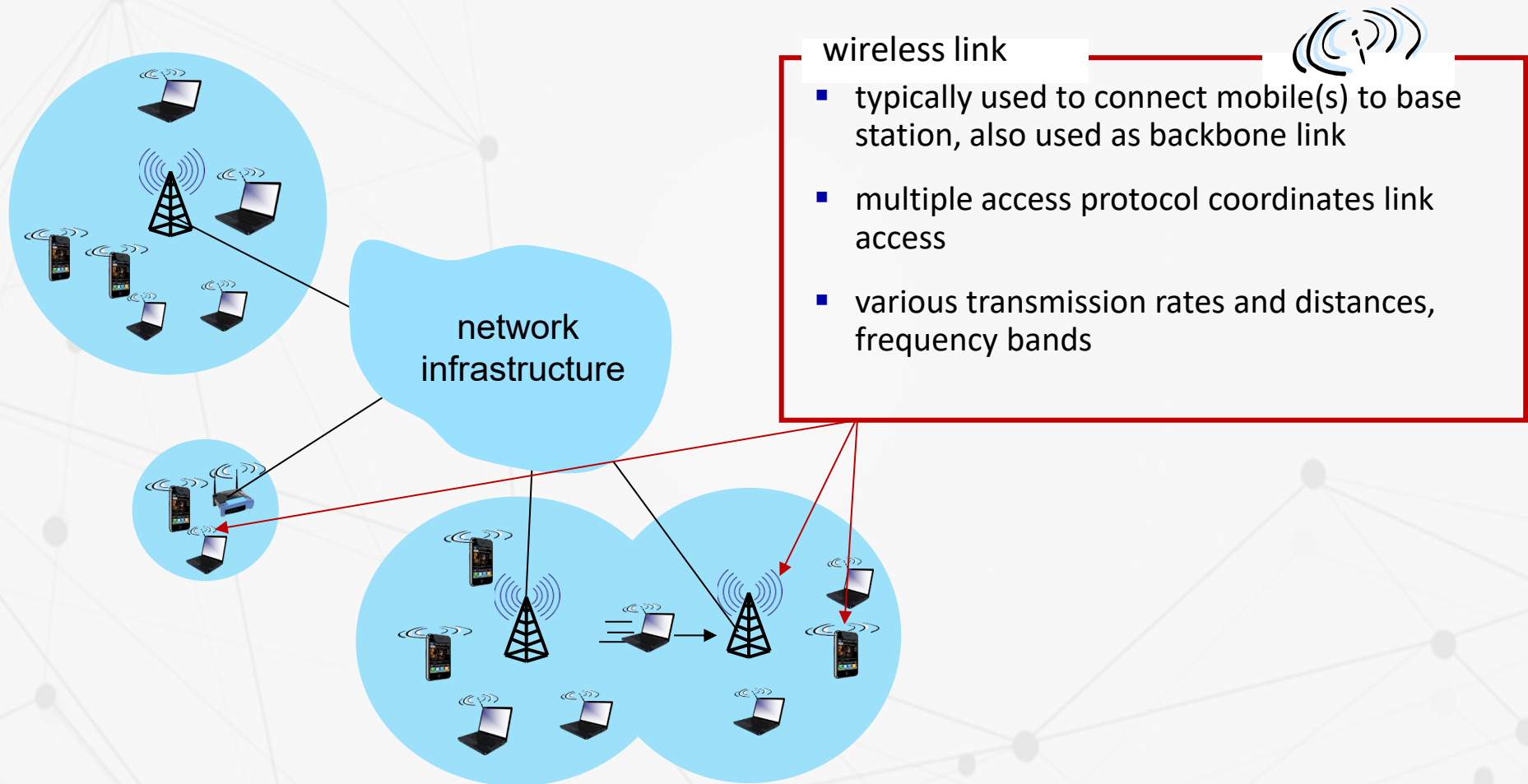
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Elements of a wireless (local area) network



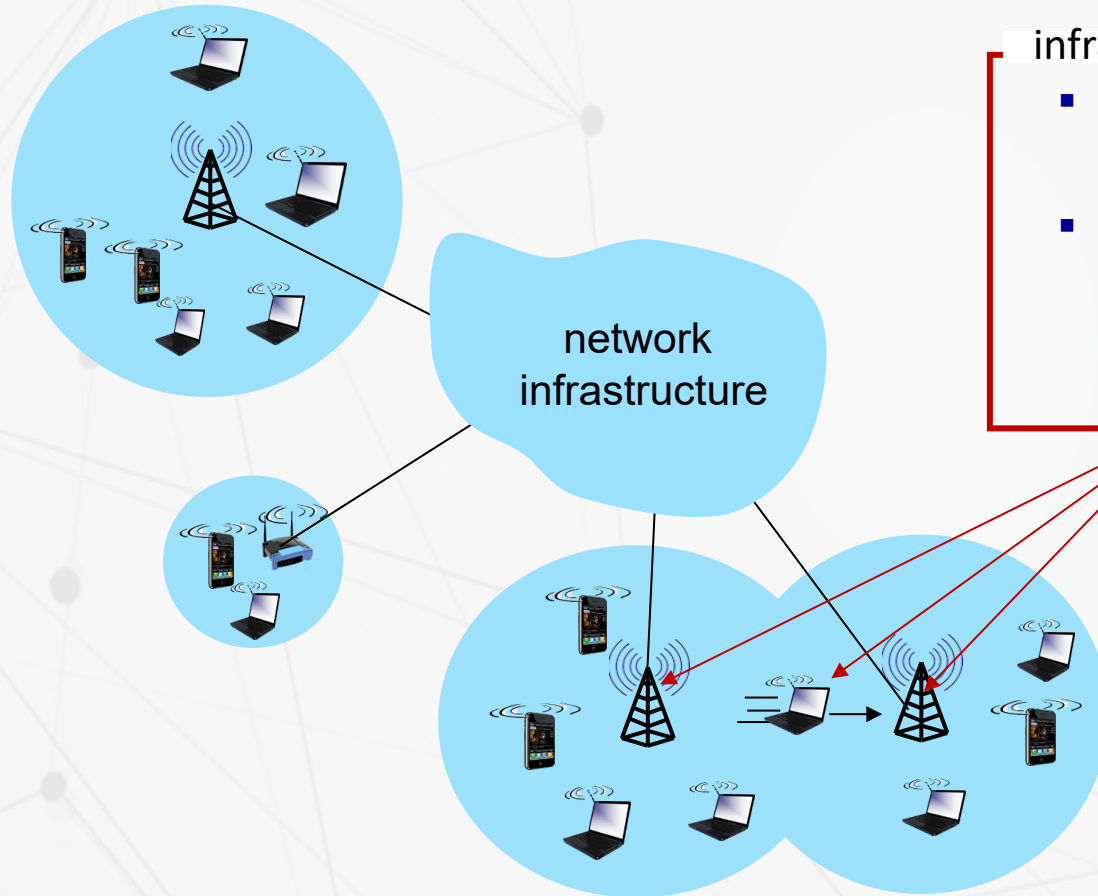
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Elements of a wireless (local area) network



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Elements of a wireless (local area) network

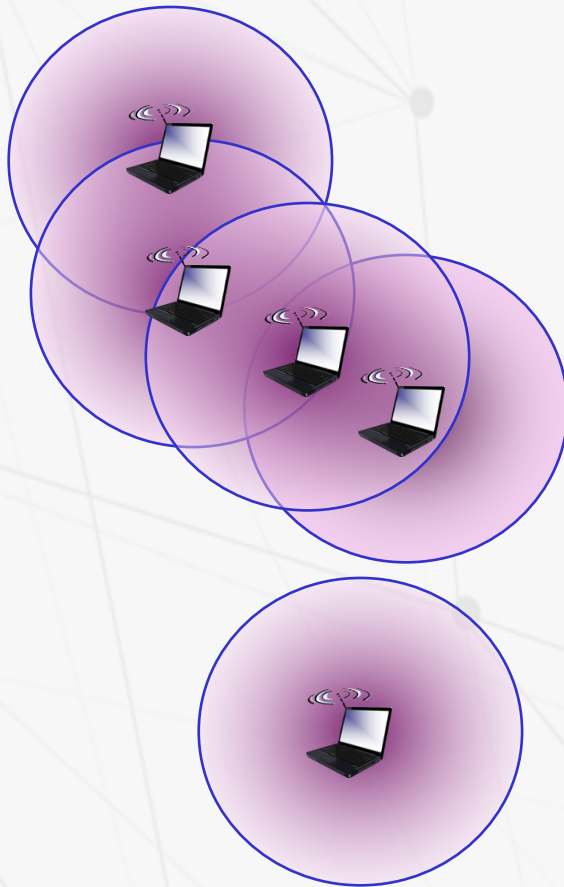


infrastructure mode

- base station connects mobiles into wired network
- handoff: mobile changes base station providing connection into wired network

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Elements of a wireless (local area) network



ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves



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Taxonomy of wireless networks

	single hop	multiple hops
infrastructure (e.g., APs)	host connects to base station (WiFi, cellular) which connects to larger Internet	host may have to relay through several wireless nodes to connect to larger Internet: <i>mesh net</i>
<i>no infrastructure</i>	no base station, no connection to larger Internet (Bluetooth, ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach other a given wireless node MANET, VANET



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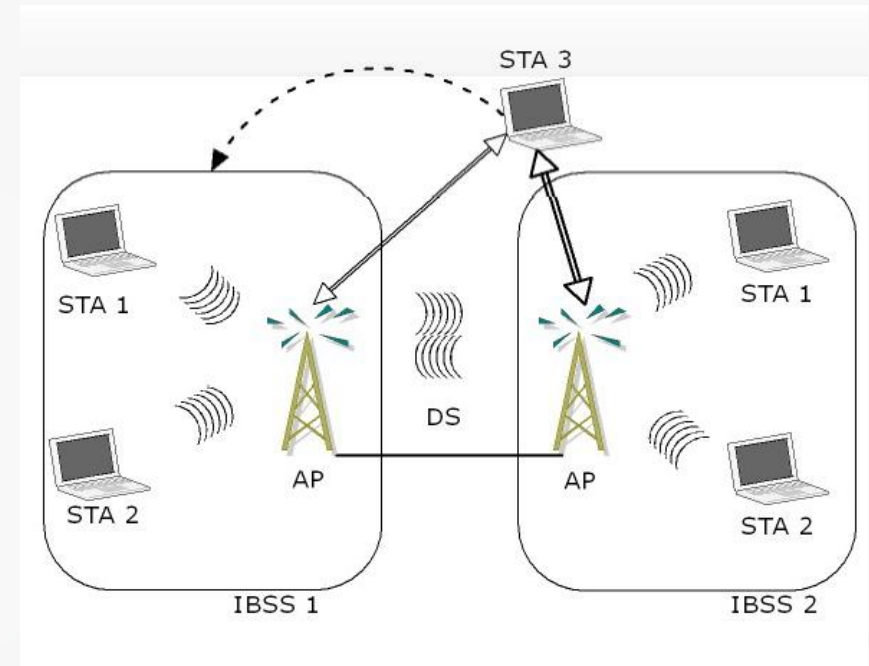
Mobility of stations

Discussion: What mobility means for Wireless Local Area Networks?

(search for related information, think and answer the following questions, 5 minutes)

Q1: How mobility" differentiates from simply being "wireless"?

Q2: Why a station seems to “move” from a BSS to another one?



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Mobility of stations

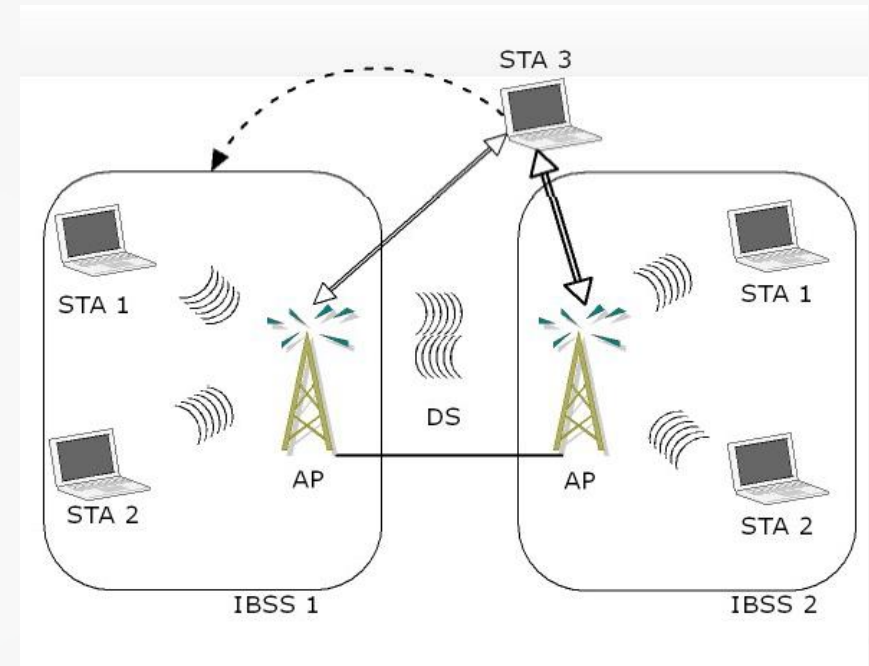
Q1: How mobility" differentiates from simply being "wireless"?

Answer: Wireless implies the absence of physical cables for connection. A device can be stationary but connected wirelessly. Mobility extends this by adding the dimension of movement.

Q2: Why a station seems to “move” from a BSS to another one?

Answer: (a) A device moves out of range of its current IBSS

(b) Manual selection by the User, disconnecting from one IBSS and then manually selecting and joining a different, from the list of available networks.



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IEEE 802.11 association



- arriving host: must **associate** with an AP
 - scans channels, listening for beacon frames containing AP's name (SSID) and MAC address
 - selects AP to associate with
 - then may perform authentication
 - then typically run Dynamic Host Configuration Protocol (DHCP) to get IP address in AP's subnet
- Note:
 - SSID (Service Set Identifier): A human-readable name (e.g., "MyHomeWi-Fi," "OfficeNetwork") associated with a BSS or ESS, allowing users to identify and connect to a network.

Source: Jim Kurose, Keith Ross, "Computer Networking: A Top-Down Approach", 9th edition Pearson, 2025.



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IEEE 802.11 Access Point (AP)



Access Point (AP)

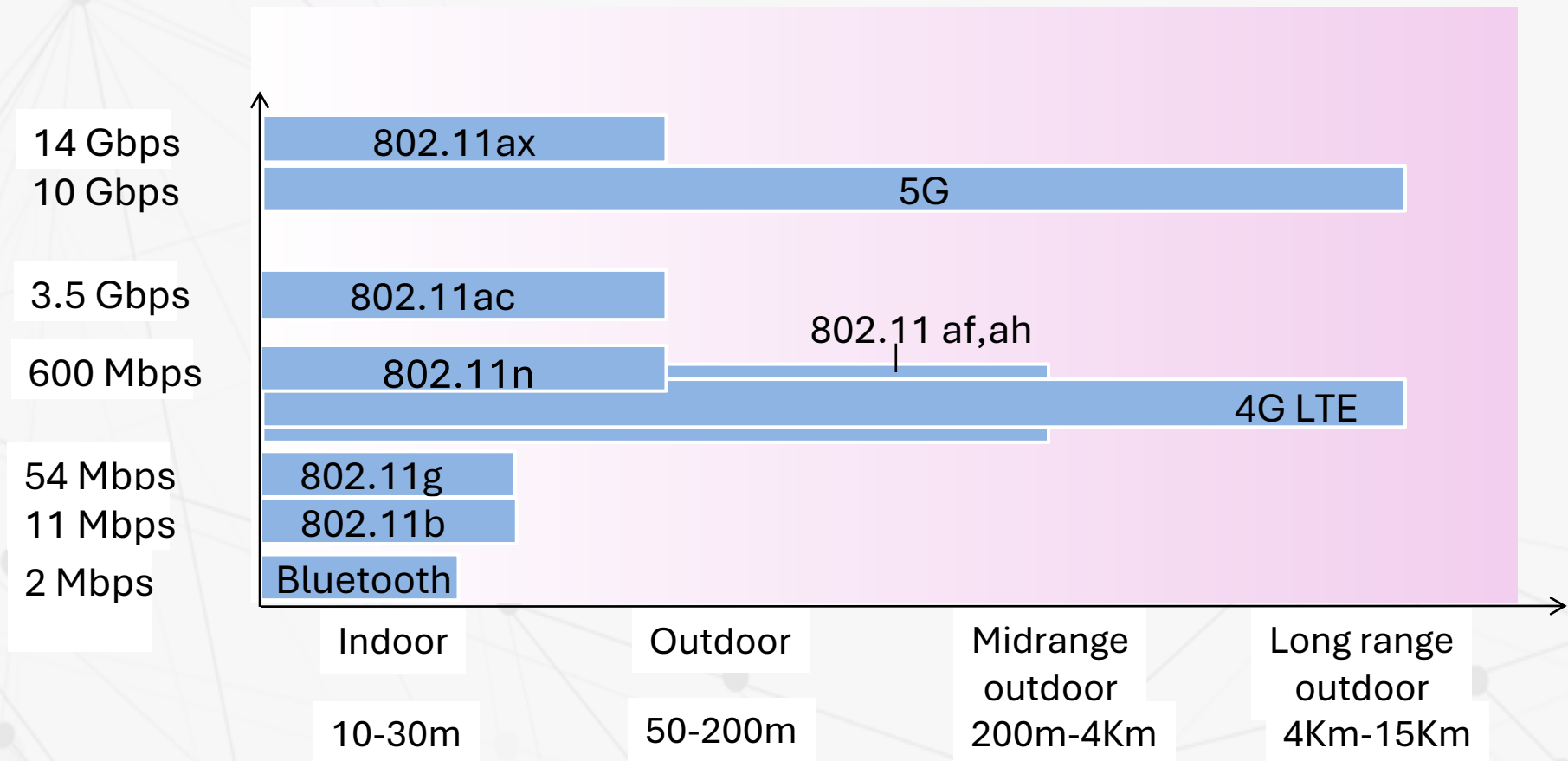
- A specialized STA that acts as a central hub and bridge between the wireless BSS and a wired Distribution System (typically an Ethernet network).
- Functionality:
 - Beaconsing: Periodically broadcasts Beacon frames to advertise the presence and capabilities of its BSS.
 - Association/disassociation: Manages requests from STAs to join or leave its BSS.
 - Authentication/deauthentication: Handles security processes for STAs.
 - Buffering: Buffers frames for power-saving STAs.
 - Frame forwarding: Forwards frames between associated STAs and between the BSS and the DS.

Source: Jim Kurose, Keith Ross, "Computer Networking: A Top-Down Approach", 9th edition Pearson, 2025.



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Characteristics of selected wireless links



Source: Jim Kurose, Keith Ross, "Computer Networking: A Top-Down Approach", 9th edition Pearson, 2025.



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IEEE 802.11 MAC sub-layer

Core principles of CSMA/CA in IEEE 802.11

How CSMA/CA in IEEE 802.11 works (in principle):

- Carrier sense: A station first "listens" to the medium to determine if it is currently idle. This involves both physical and virtual carrier sense.
- Wait and backoff: If the medium is busy, or after a transmission, the station waits for a specific InterFrame Space (IFS) and then enters a random backoff process. The various IFSs provide varying priority for different types of frames.
- Transmit: If the medium remains idle after the backoff, the station transmits its frame.
- Acknowledgement (ACK): The receiving station sends an immediate ACK frame upon successful reception.
- Retransmission: If the sender does not receive an ACK, it assumes a collision or loss and retransmits the frame after another backoff.



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Physical carrier sensing in IEEE 802.11

Physical carrier sensing (Clear Channel Assessment - CCA):

- Mechanism: Performed at the Physical layer (PHY). A station literally listens to the radio channel for signals.
- Detection methods:
 - Energy Detect (ED): Checks if the received signal strength (RSSI) on the channel is above a certain threshold. If so, the medium is considered busy.
 - Preamble Detect (PD): Detects an IEEE 802.11 preamble (the start of an IEEE 802.11 frame). This is more precise as it indicates an actual IEEE 802.11 transmission, not just general noise.
 - Limitation: Prone to the hidden station problem; a station might not hear another transmitting station, leading to collisions.



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Virtual carrier sensing in IEEE 802.11

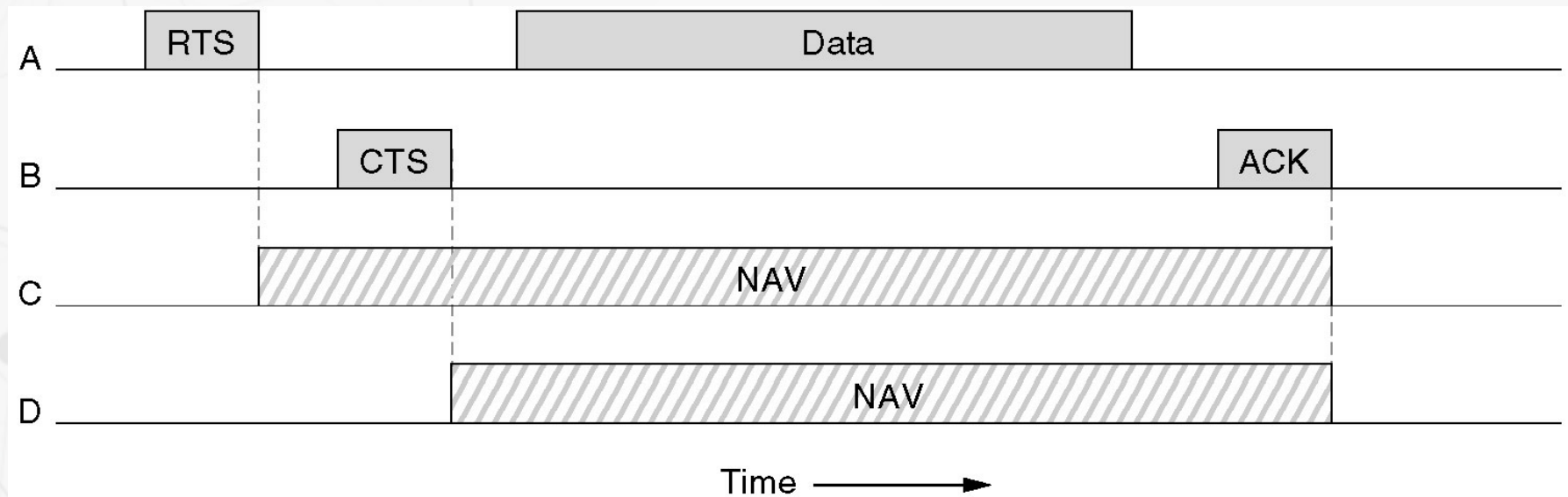
Virtual carrier sensing (Network Allocation Vector - NAV):

- It's a timer maintained by each station, indicating the duration for which the medium is expected to be busy. It effectively addresses the hidden station problem.
- Setting and utilizing NAV:
 - Duration/ID field: Almost all IEEE 802.11 MAC frames (data, management, control) contain a "Duration/ID" field. When a station receives any frame, it reads this field and updates its NAV if the new value is greater than its current NAV.
 - NAV update: All stations within range hearing any frame update their NAV, effectively "reserving" the medium for the duration specified (even for stations that cannot hear the actual data transmission).
 - For data frames and acknowledgement (ACK) frames, this field typically indicates the remaining time until the current transmission sequence (e.g., Data + ACK) is complete. For control frames (RTS/CTS) it plays a crucial role in medium reservation.
 - Principle: A station can only transmit if both its physical carrier sense indicates the medium is idle and its NAV has counted down to zero.



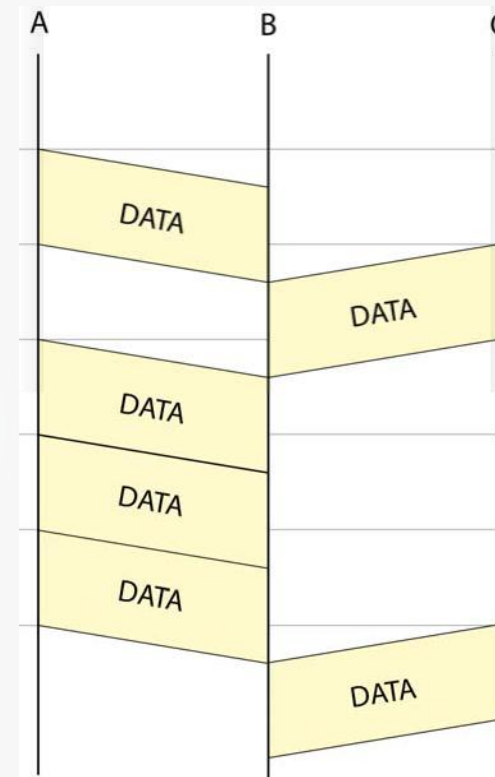
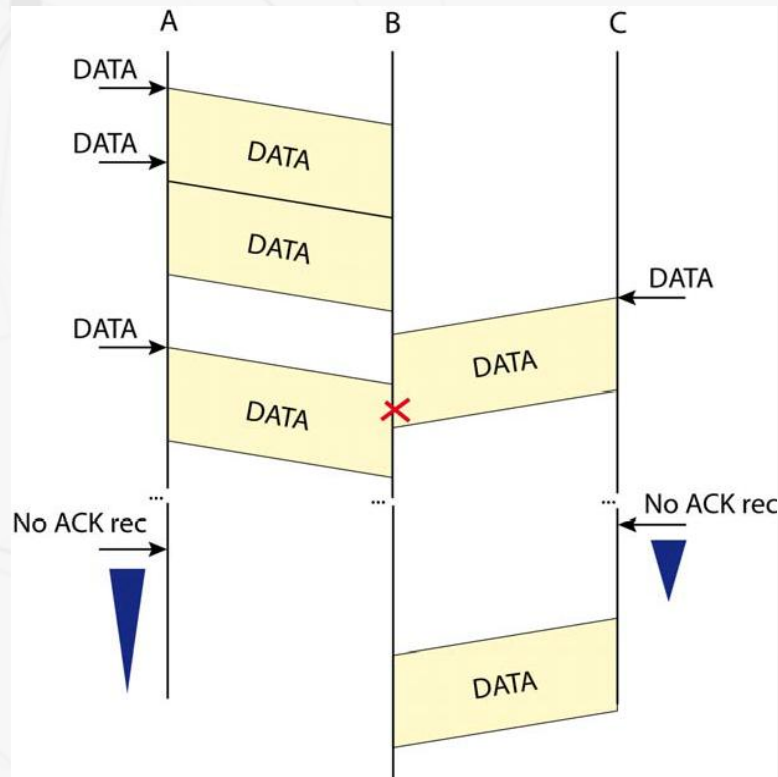
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Virtual channel sensing in IEEE 802.11



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Performance (throughput)



Ideal situation: If we could schedule the transmissions perfectly, such that there is neither an overlap and nor a gap between the messages, we could achieve the maximum throughput.



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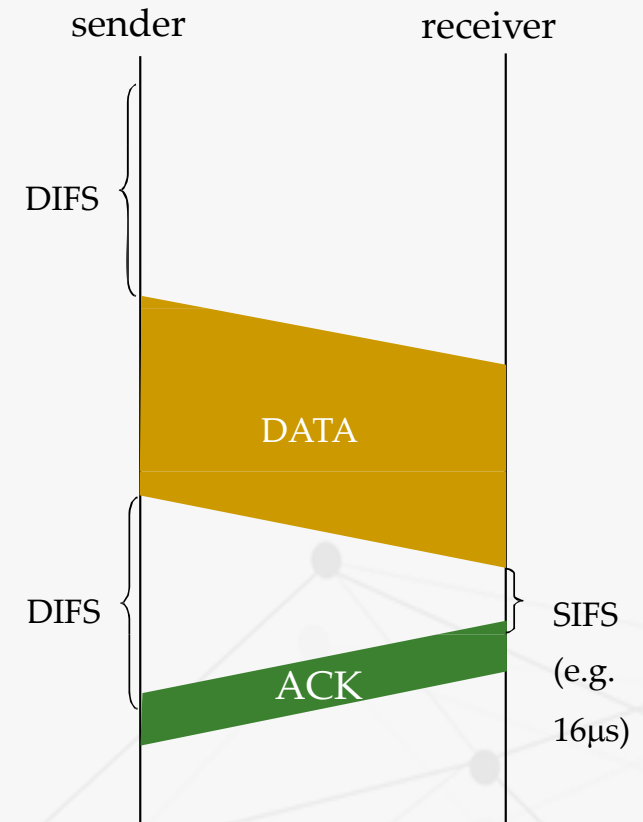
CSMA in IEEE 802.11 (Basic Access method)

IEEE 802.11 sender (when trying to send)

- Step 1: If sense channel idle for DIFS then transmit frame
- Step 2: If sense channel busy then:
 - 2.1 Start random backoff timer, count down the backoff timer only while channel idle and freeze when busy (resumes if idle for DIFS)
 - 2.2 Transmit when timer expires
- Step 3: If ACK received, wait for DIFS, then go to step 2.1
- Step 4: If no ACK received, double backoff range, select time randomly from range, repeat step 2

IEEE 802.11 receiver

- If frame received successfully then return ACK after SIFS interval (ACK needed due to hidden terminal problem)



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CSMA in IEEE 802.11

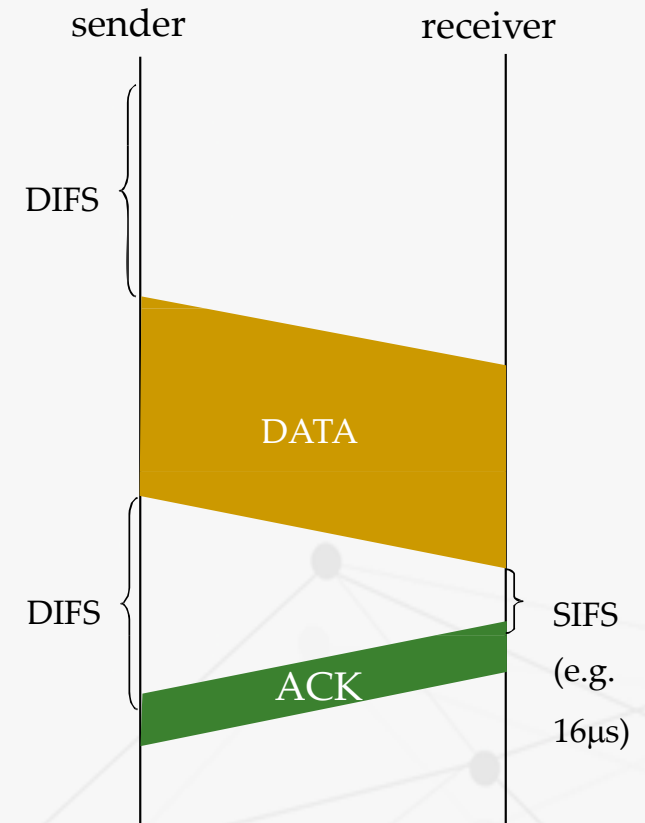
Discussion: How CSMA really works in IEEE 802.11?

(search for related information, think and answer the following questions, 5 minutes)

Q1: Why does the station transmit when timer expires?

Q2: If ACK received, why do we need to wait for DIFS?

Q3: Why If no ACK received, why do we need to double backoff range, select time randomly from range and repeat step 2?



IEEE 802.11 MAC sub-layer

Location dependent carrier sensing

Hidden nodes:

- A hidden node is one that is within the range of the intended destination but out of range of the sender

Exposed nodes:

- An exposed node is one that is within the range of the sender but out of range of the destination

Capture:

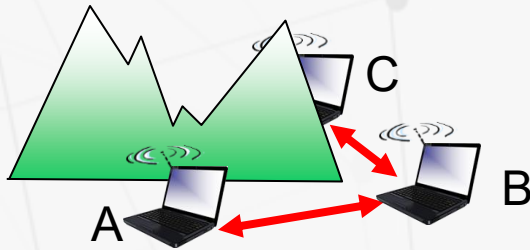
- This event occurs when a receiver can correctly receive a transmission from one of two (or more) simultaneous transmissions, all within its range, because the signal strength of the correctly received signal is much higher than strength of the other signals



IEEE 802.11 MAC sub-layer

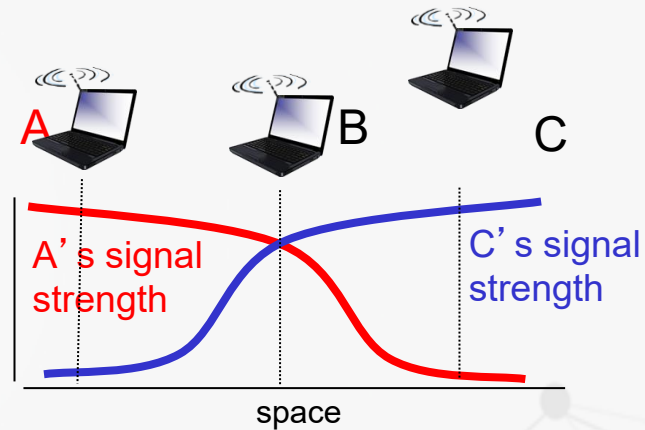
Wireless link characteristics: Hidden nodes

Hidden node problem



- B, A hear each other
- B, C hear each other
- A, C can not hear each other means A, C unaware of their interference at B

Attenuation also causes “hidden nodes”



- B, A hear each other
- B, C hear each other
- A, C can not hear each other interfering at B

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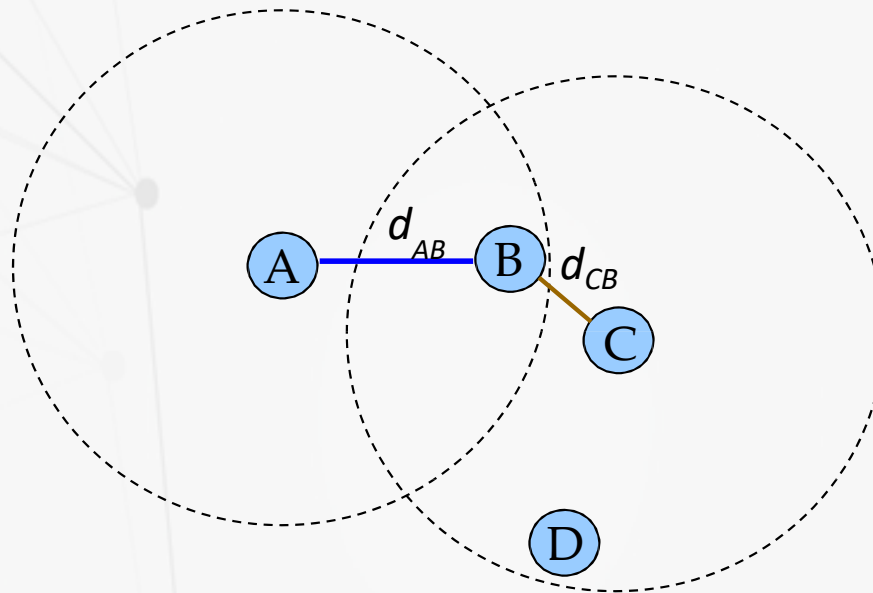
Wireless link characteristics: Exposed nodes



- B is transmitting to A
- C wants to transmit to D
- C senses transmission & declines even if its transmission to D will not cause any collision to A

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Wireless link characteristics: Capture problem

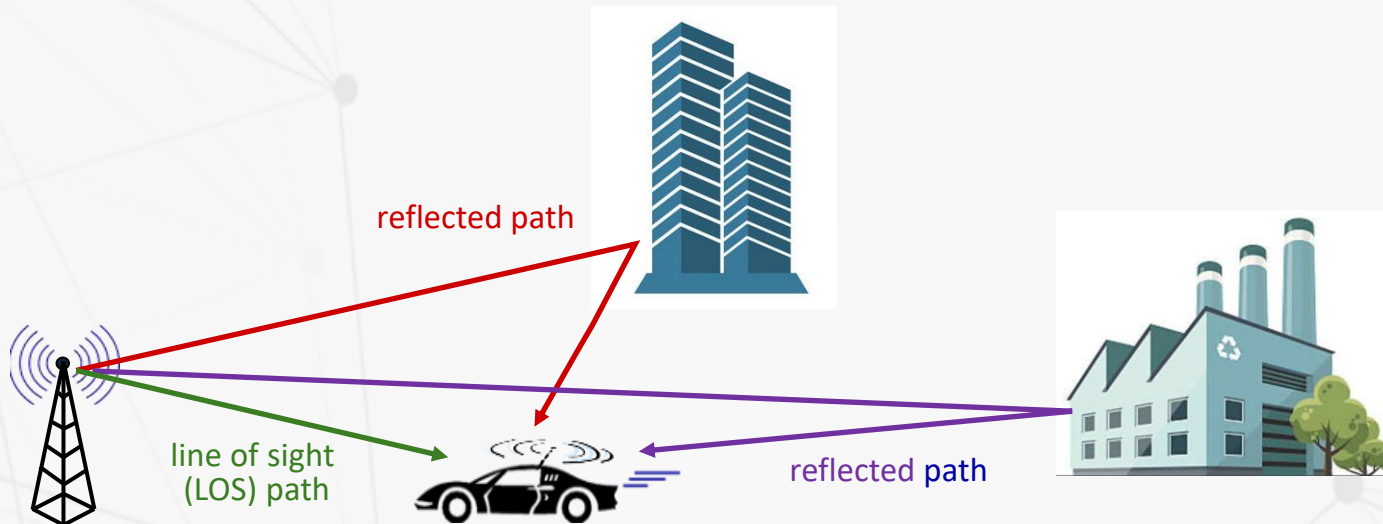


- If A and C transmit simultaneously to B then the signal power of C, received at B, is higher than the one from A and there is a good probability that C's signal can be correctly decoded in the presence of A's transmission.
- This capture of C's signal can improve protocol performance, but it results in unfair sharing of the channel with preference given to nodes closer to the receiver.



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Wireless link characteristics: multipath



Multipath propagation: radio signal reflects off objects ground, built environment, arriving at destination at slightly different times.



IEEE 802.11 MAC sub-layer

Collision avoidance through exchange of RTS-CTS

RTS/CTS mechanism

- Collision for long packets is wasteful
- RTS (Request to Send): Request to reserve channel to send long packet w/o collisions
- CTS (Clear to Send): Approve RTS
- Optional mechanism

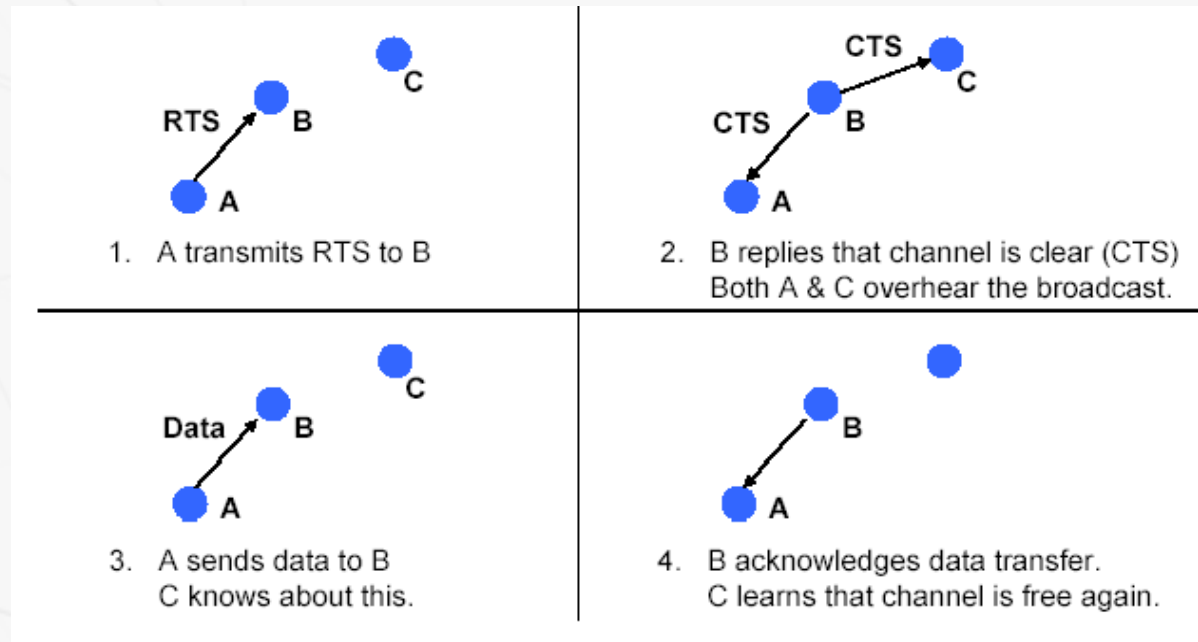
idea: sender “reserves” channel use for data frames using small reservation packets

- ☐ sender first transmits *short* request-to-send (RTS) packet to receiver using CSMA
- ☐ BS broadcasts clear-to-send CTS in response to RTS
- ☐ CTS heard by all nodes
 - sender transmits data frame
 - other stations defer transmission



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Collision avoidance through exchange of RTS-CTS



RTS-CTS handshake used to resolve hidden terminal problem and shorten the duration of collisions

- RTS = Ready to Send; transmitted by sender to inform receiver of available data
- CTS = Clear to Send; broadcasted by receiver to inform all neighbors that channel will be occupied
- ACK = Acknowledgement; broadcasted by receiver to inform all neighbors that channel is free again

IEEE 802.11 MAC sub-layer

Collision avoidance through exchange of RTS-CTS

Discussion: Are there any drawbacks of RTS/CTS mechanism?

(search for related information, think and answer the following questions, 10 minutes)

Q1: Do we have collisions with the use of RTS/CTS?

Q2: Is there any overhead that RTS/CTS adds?

Q3: The RTS/CTS frames are always transmitted using the control rate that is lower than the data rate. Do you revise your previous reply regarding overhead?

Q4: Could you propose some ideas for the optimum use of RTS/CTS? When it is beneficial to be employed?



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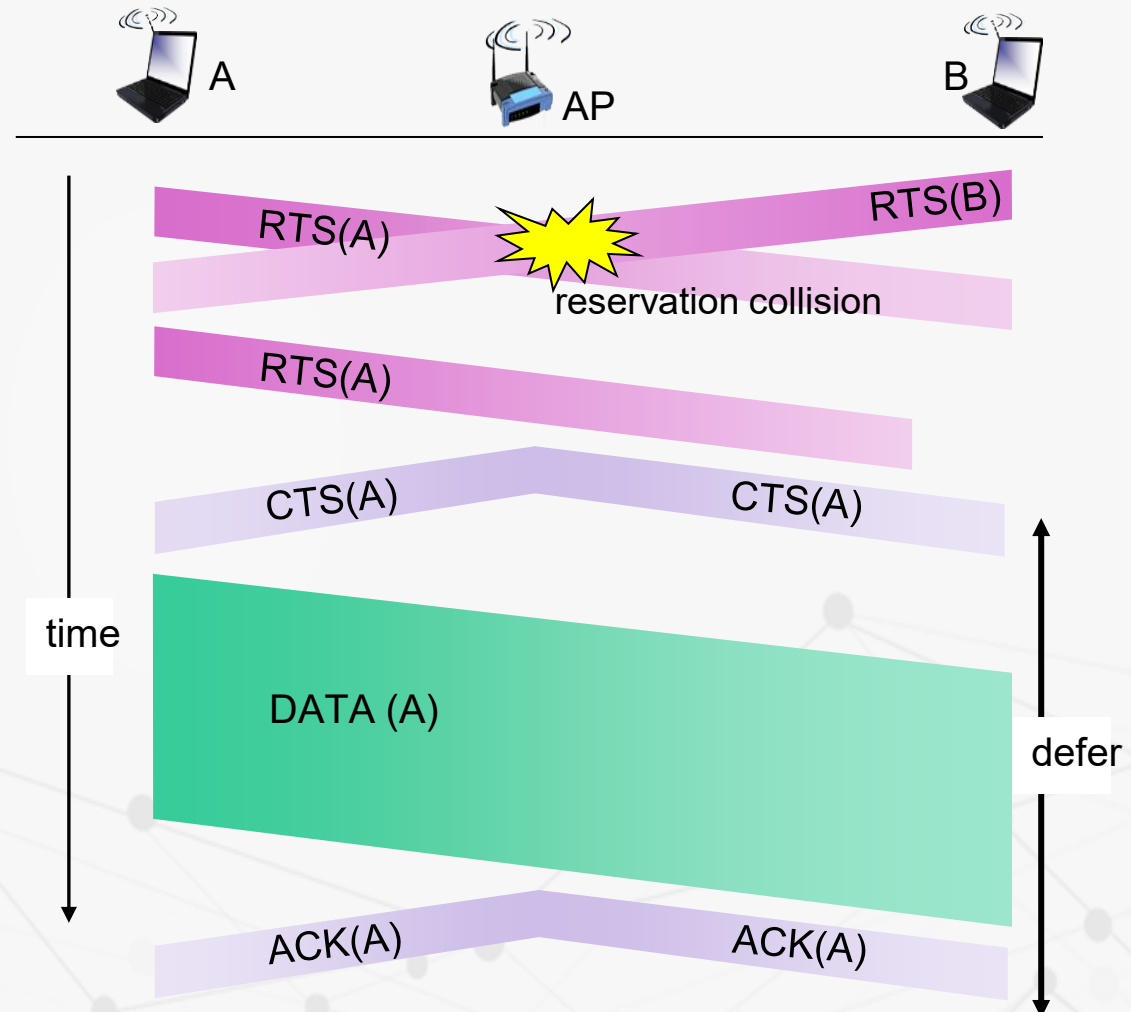
Collision avoidance through exchange of RTS-CTS

Q1: Do we have collisions with the use of RTS/CTS?

Answer:

The RTS/CTS handshake reduces number of collisions (not avoids them completely).

RTSs may still collide with each other (but they're short), thus, collision duration is reduced.



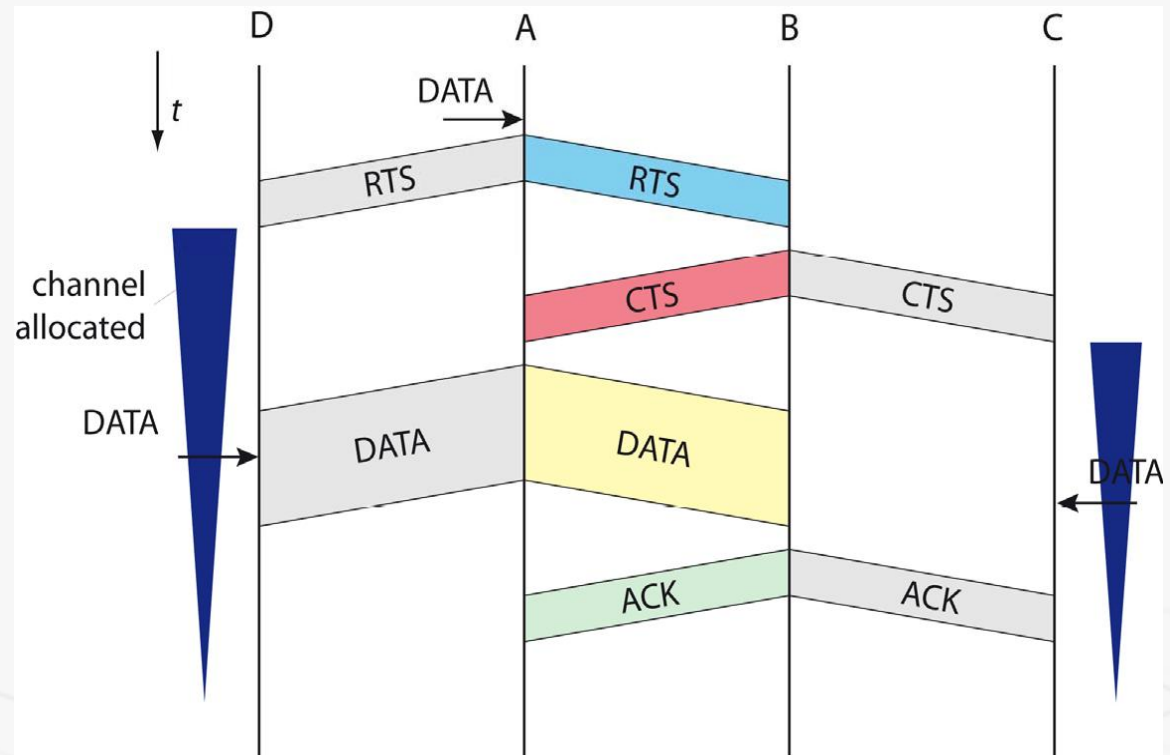
IEEE 802.11 MAC sub-layer

Collision avoidance through exchange of RTS-CTS

Q2: Is there any overhead that RTS/CTS adds?

Q3: The RTS/CTS frames are always transmitted using the control rate that is lower than the data rate. Do you revise your previous reply regarding overhead?

Answer: The RTS/CTS handshake adds significant signaling overhead, especially if data messages are short.



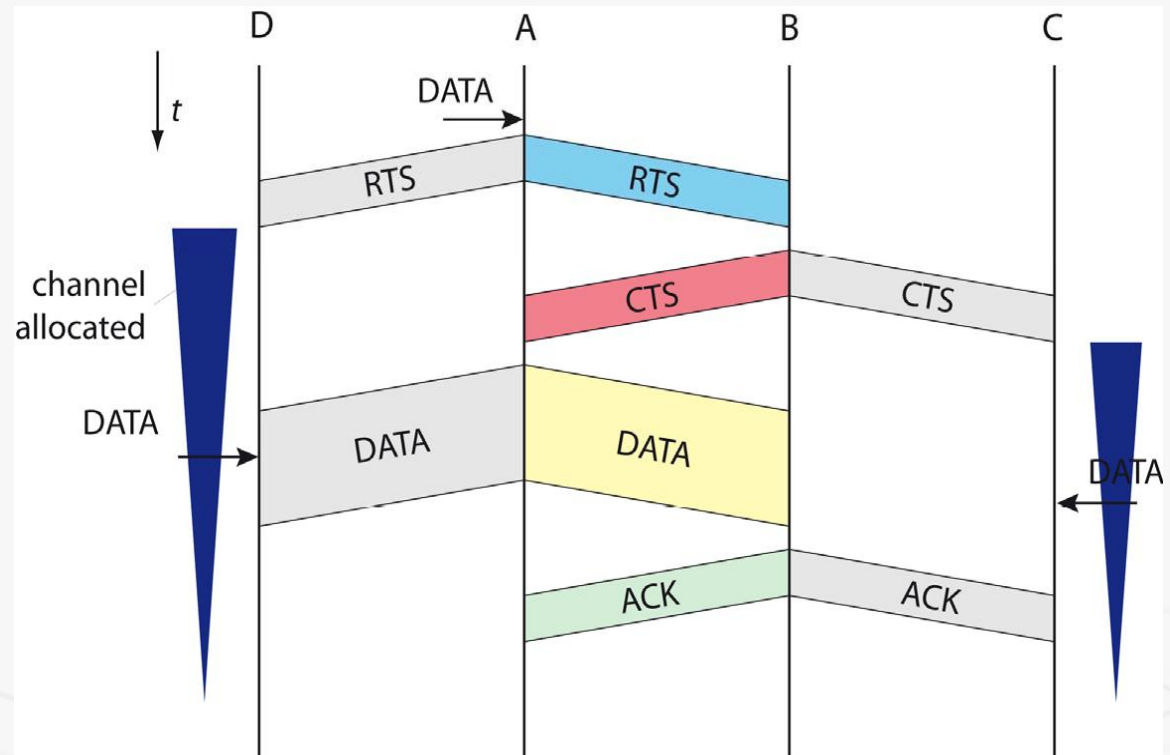
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Collision avoidance through exchange of RTS-CTS

Q4: Could you propose some ideas for the optimum use of RTS/CTS? When it is beneficial to be employed?

Answer: The RTS/CTS depends on the size of data frames and the frequency of collisions (potential presence of hidden stations or dense environments).

Can we tell if they are hidden stations in our network? Can we find the number of stations in the network?



Note: Stations typically use an RTS Threshold parameter (configurable on APs and wireless clients). If the size of the data frame to be transmitted exceeds this threshold, the RTS/CTS handshake is initiated.



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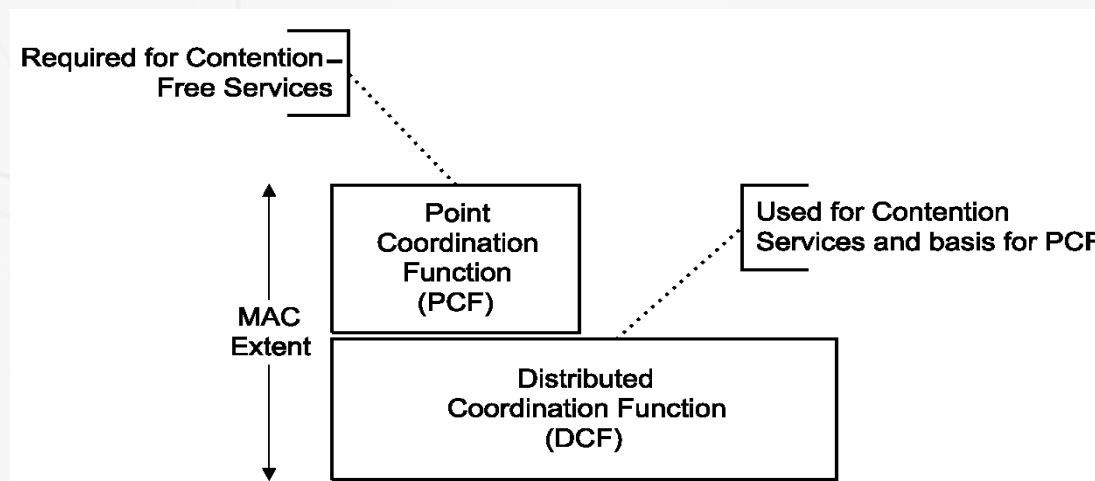
Access methods in IEEE 802.11 MAC sub-layer

DCF (Distributed Coordination Function)

- Ad hoc and infrastructure mode
- Contention based distributed system
- Mandatory

PCF (Point Coordination Function)

- Infrastructure mode, optional
- Contention free centralized system
- Polling to coordinate senders, e.g. to ensure QoS
- For real time service



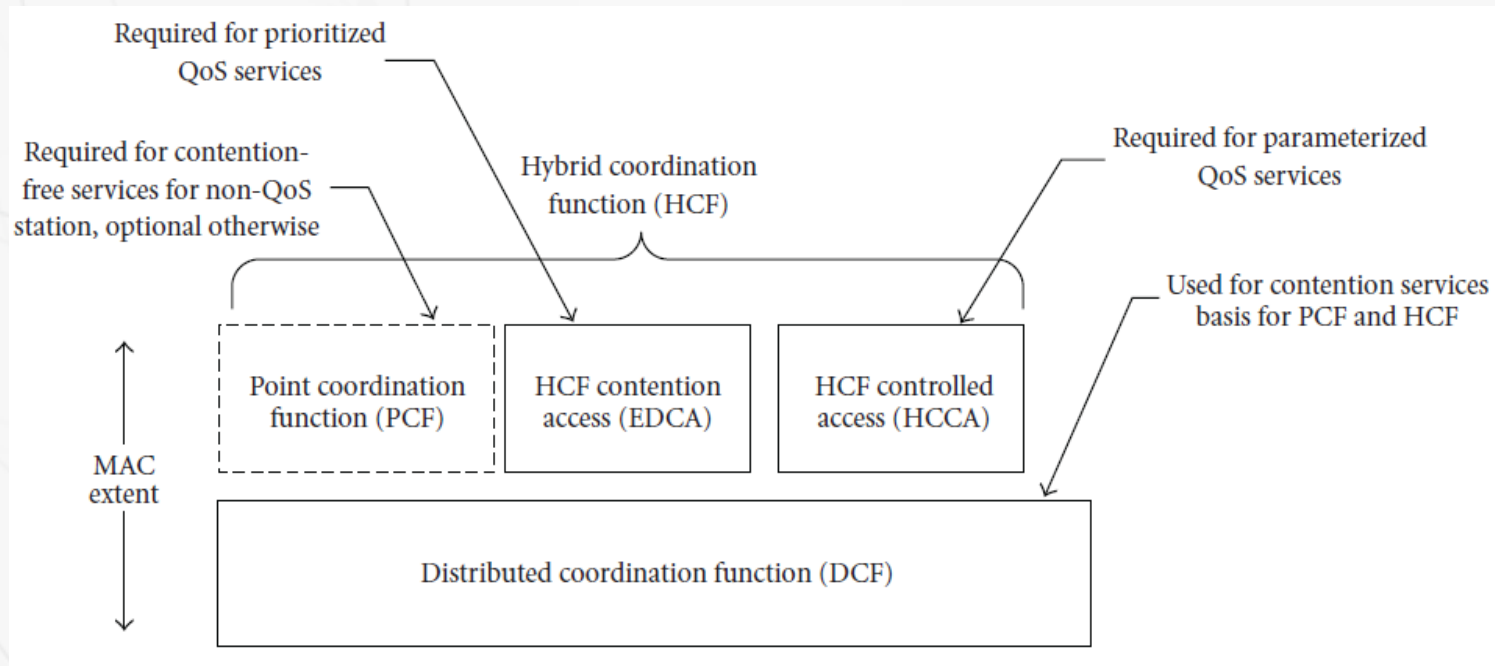
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Access methods in IEEE 802.11 MAC sub-layer

Hybrid Coordination Function (HCF):

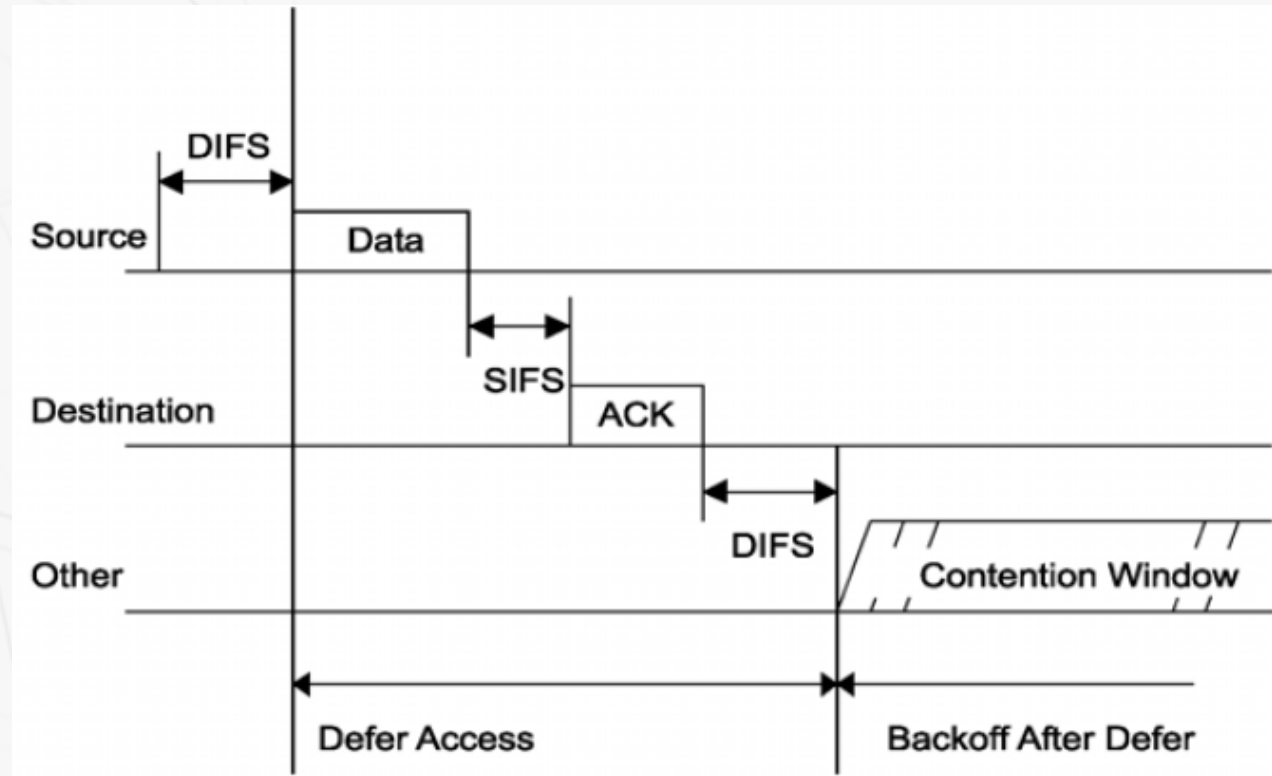
The QoS-aware successor, including:

- Enhanced Distributed Channel Access (EDCA): Contention-based with priorities.
- HCF Controlled Channel Access (HCCA): Centralized, scheduled access for guaranteed QoS.



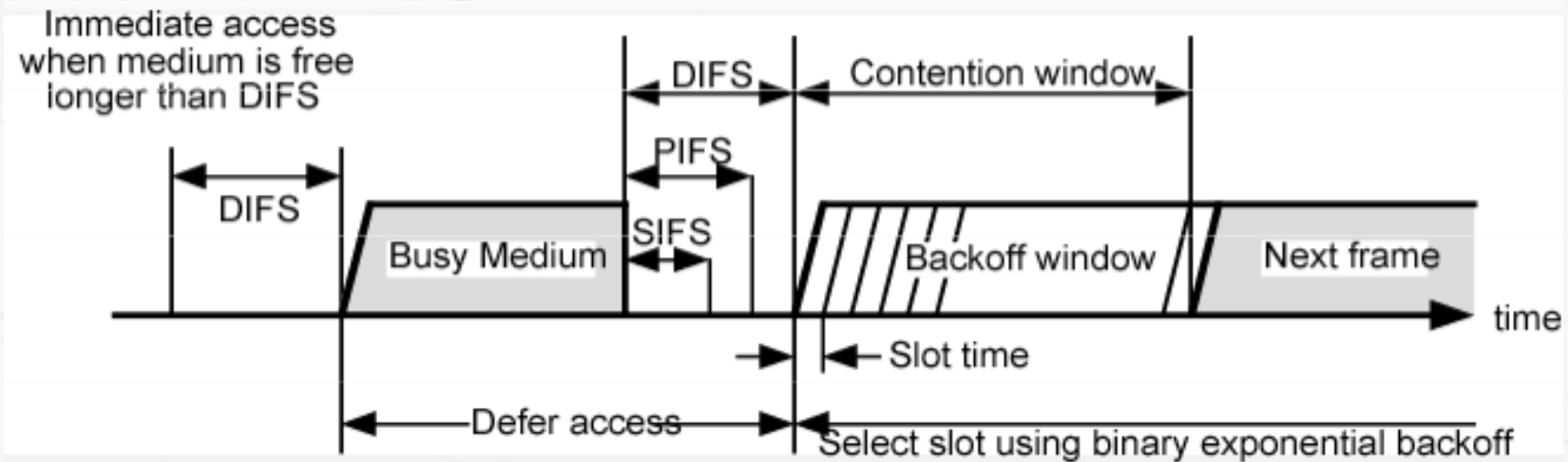
IEEE 802.11 MAC sub-layer

Distributed Coordination Function (DCF) - Basic Access method



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Distributed Coordination Function (DCF) - Basic Access method



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Point Coordination Function (PCF)

PCF is an optional access method in the original IEEE 802.11 standard that operates under the control of a Point Coordinator (PC), which is typically the Access Point (AP).

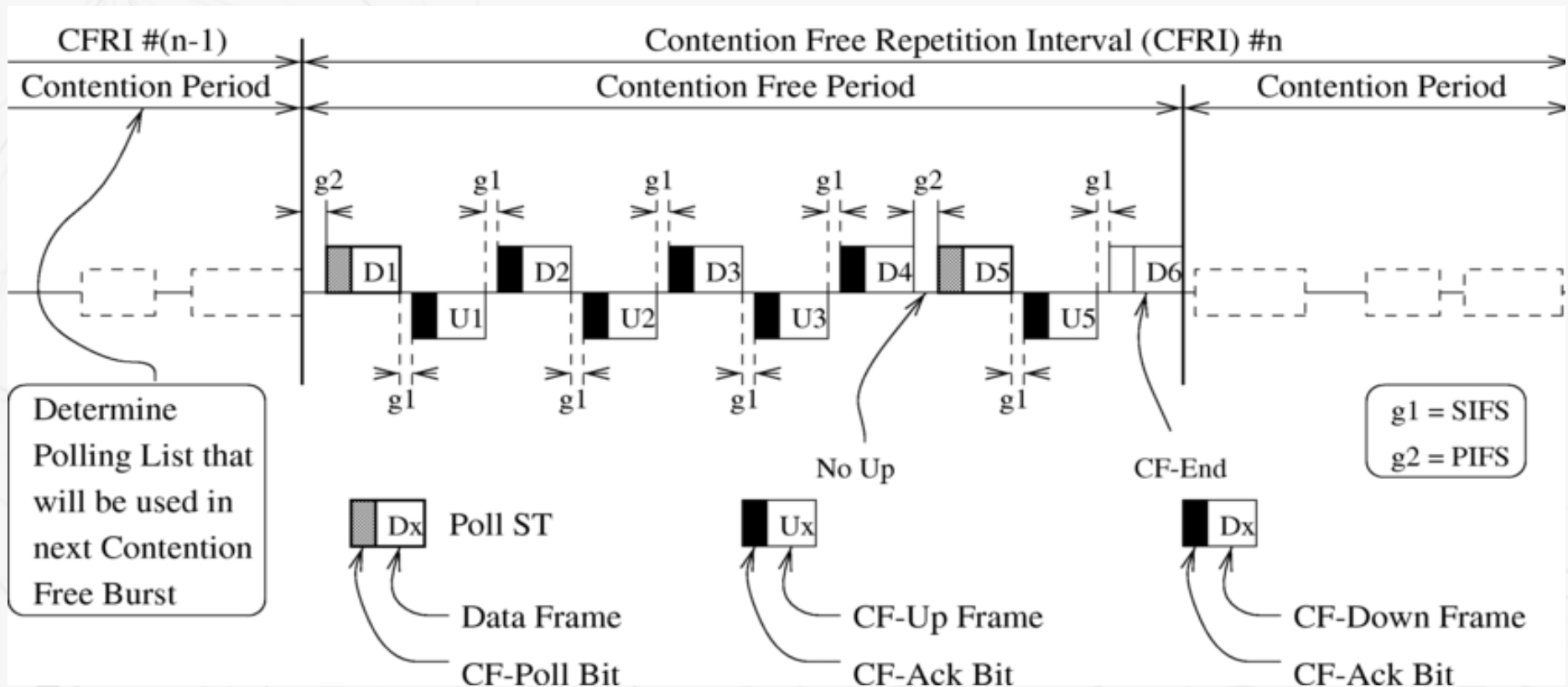
Unlike the contention-based DCF, PCF uses a polling mechanism. The PC (AP) grants permission to stations to transmit, eliminating contention during the Contention-Free Period (CFP).

Operation during CFP:

- The AP begins a CFP by sending a special Beacon frame that indicates the start and duration of the CFP.
- During the CFP, the AP uses PIFS to gain priority over DCF stations.
- The AP polls stations in a round-robin fashion using CF-Poll frames.
- A polled station can transmit one data frame (or a combined data/ACK frame) after receiving a CF-Poll, without contention.
- If a polled station has no data, it sends a Null frame or CF-ACK.
- At the end of the CFP (or if no response to polling), the AP sends a CF-End frame to revert to DCF mode.



Point Coordination Function (PCF)



IEEE 802.11 MAC sub-layer

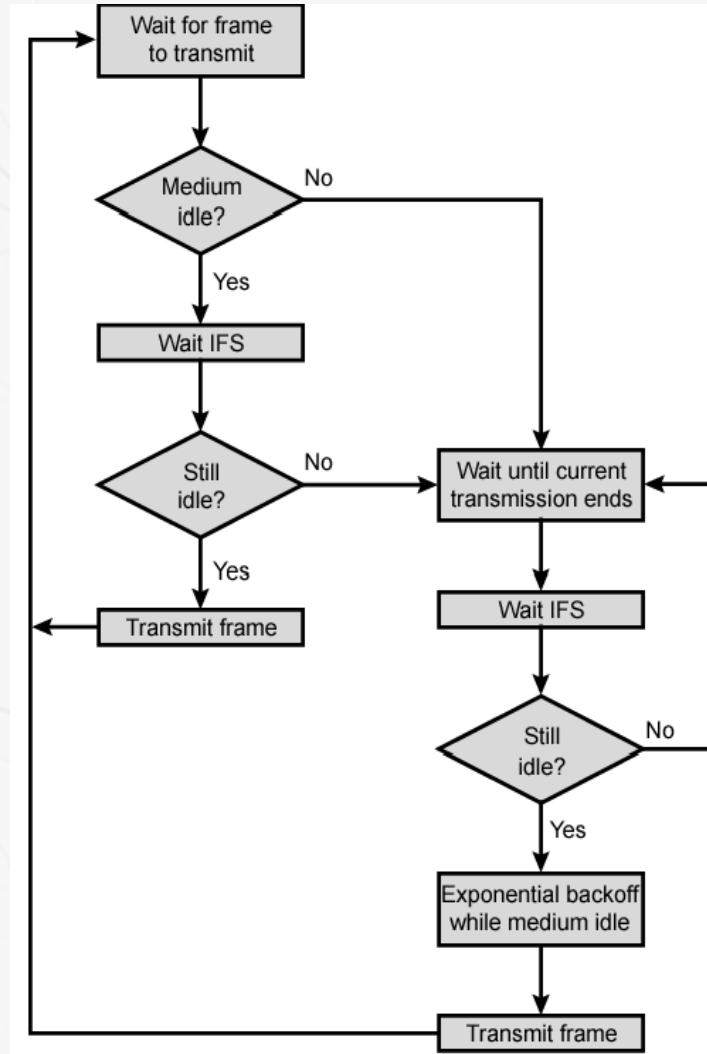
Contents

- Carrier Sense Multiple Access basics
- IEEE 802.11 architecture
- Physical and virtual channel sensing
- Access methods in IEEE 802.11 MAC sub-layer
- IEEE 802.11 MAC basics



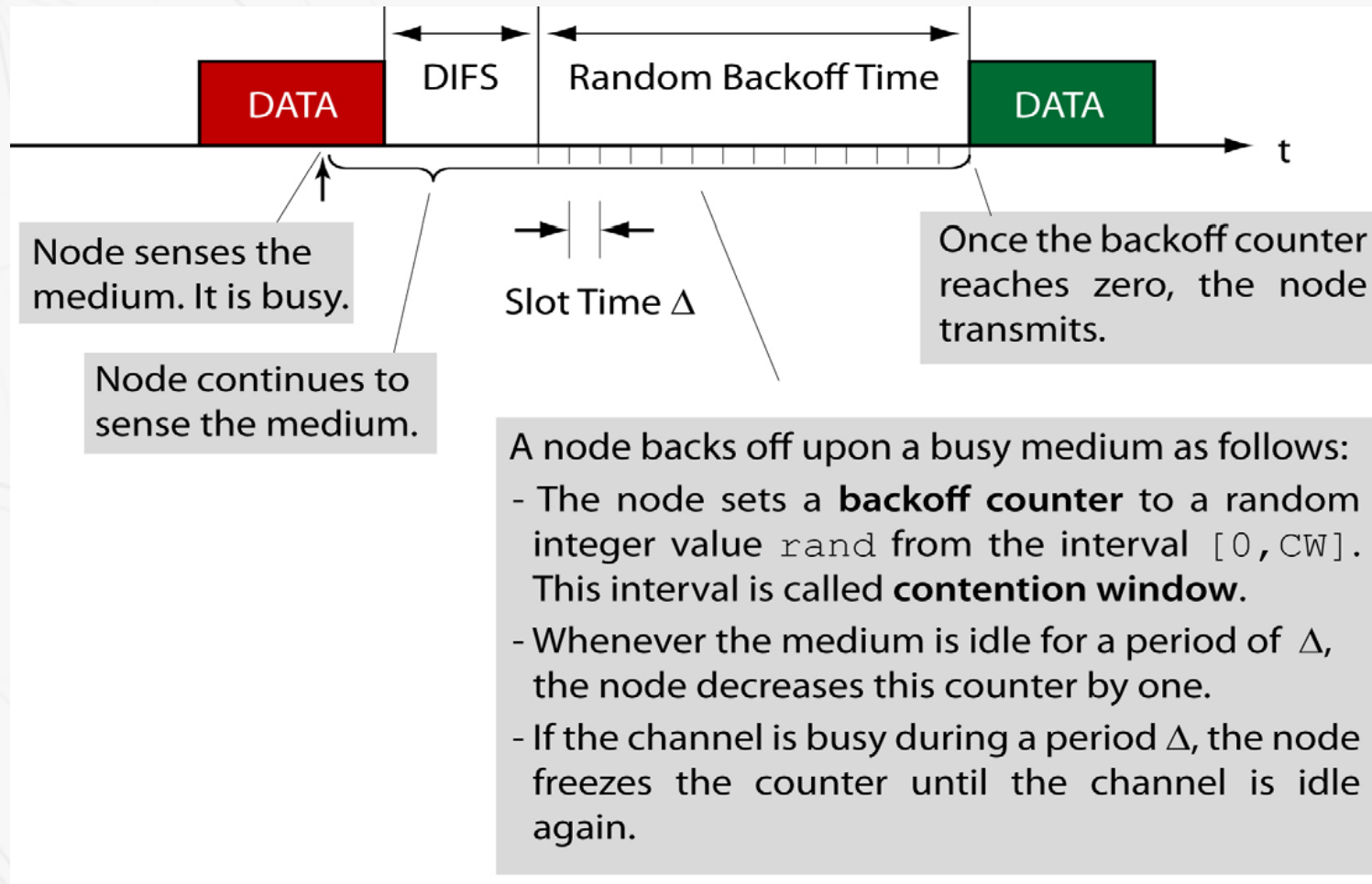
IEEE 802.11 MAC sub-layer

Transmission procedure



IEEE 802.11 MAC sub-layer

Busy channel and backoff



IEEE 802.11 MAC sub-layer

Backoff procedure

To avoid collision, backoff stage is considered to determine the contention window (CW). A station picks up a random number in the current contention window.

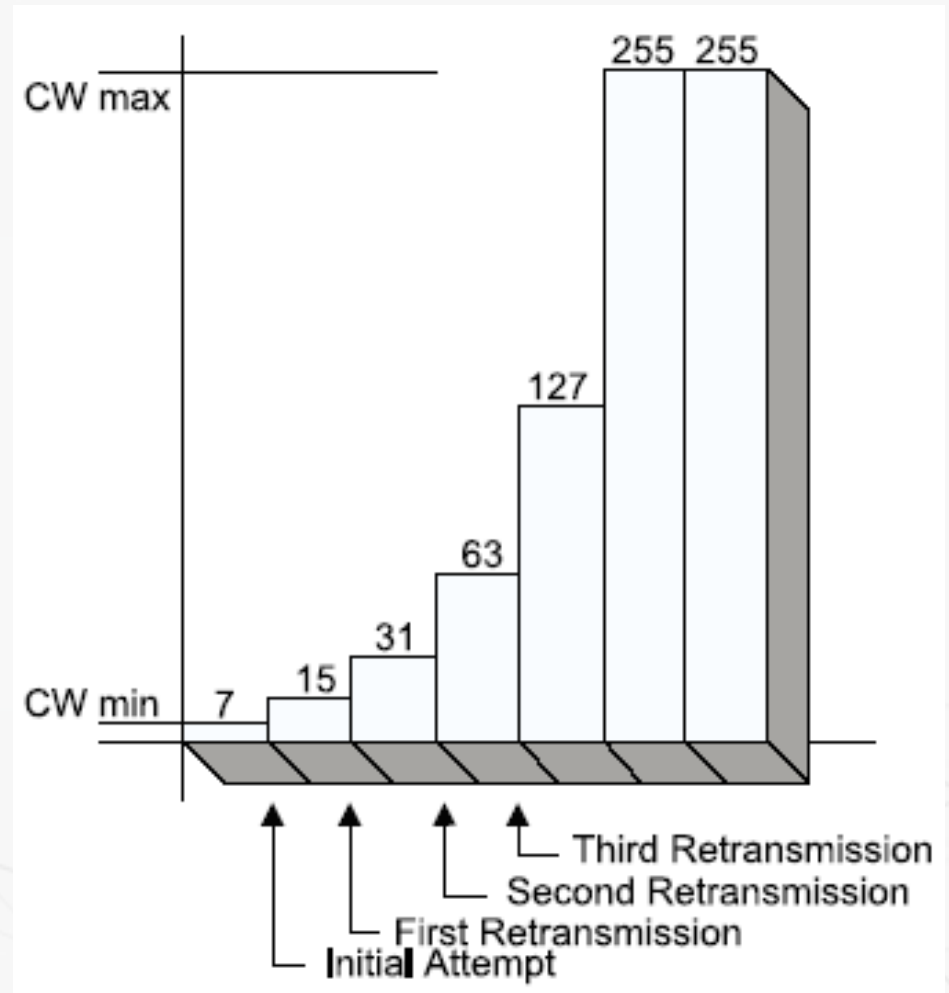
The Binary Exponential Backoff (BEB) scheme is used as follows:

$$CW = 2^k(CW_{min} + 1) - 1,$$

where k = the backoff stage (0,...K)

Upon success (ACK received):

$$CW = CW_{min}$$



IEEE 802.11 MAC sub-layer

Backoff procedure

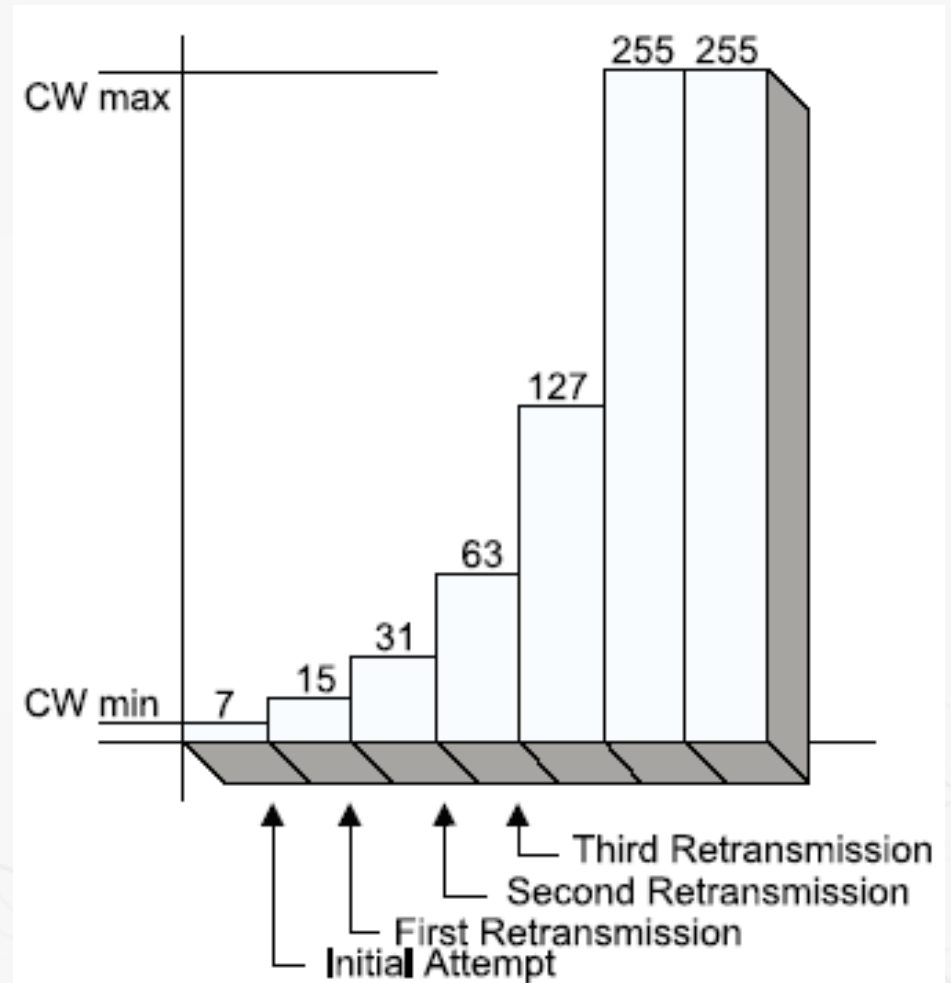
Discussion: How backoff procedure works in IEEE 802.11?

(search for related information, think and answer the following questions, 5 minutes)

Q1: Why we double the contention window after every failed transmission?

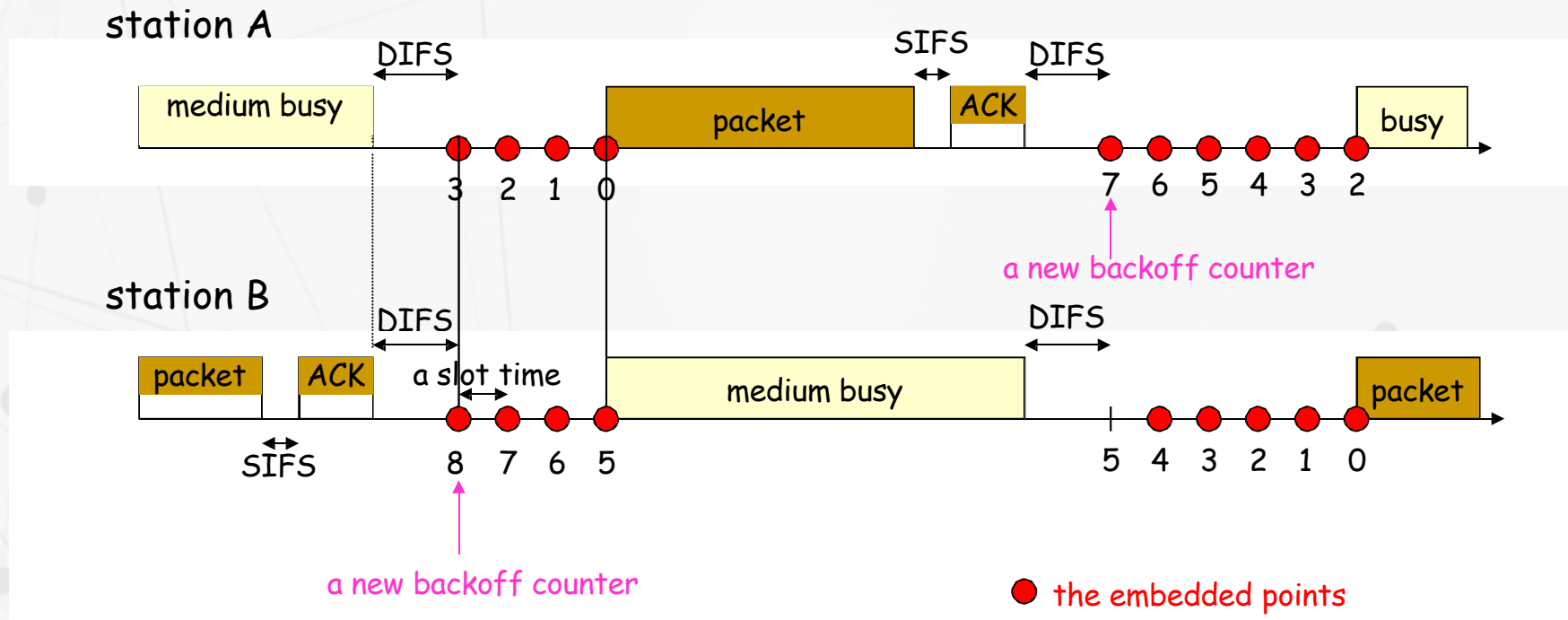
Q2: Why do we keep constant the contention window after reaching the maximum value (CW_{max})?

Q3: Why do we reset the contention window to CW_{min} after a successful transmission?



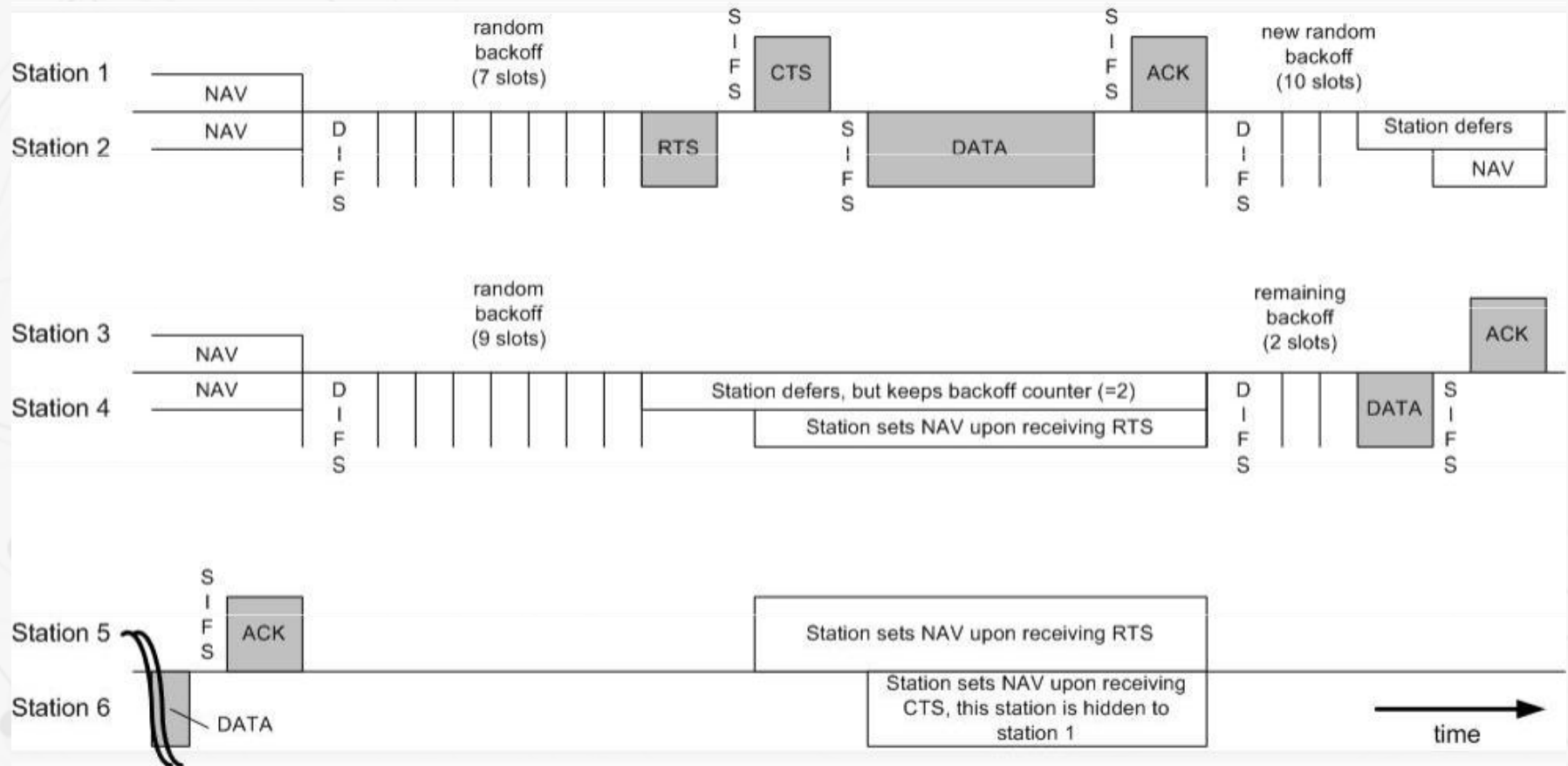
IEEE 802.11 MAC sub-layer

Example of Basic Access and backoff



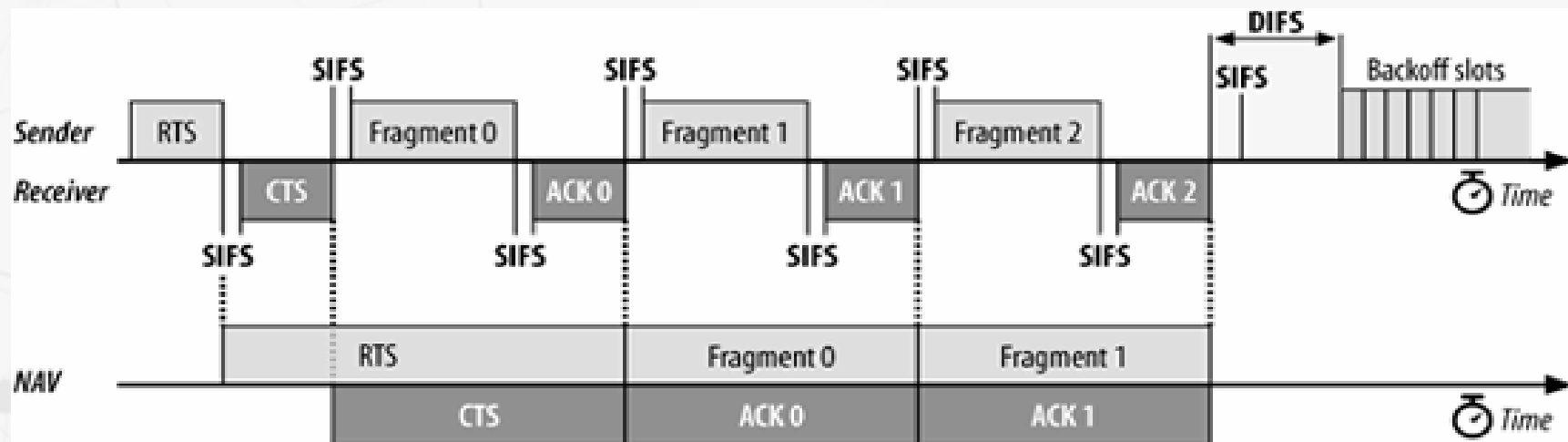
IEEE 802.11 MAC sub-layer

Example of RTS/CTS and backoff



IEEE 802.11 MAC sub-layer

Fragmentation



IEEE 802.11 MAC sub-layer

Retry limit

Stations maintain a Station Short Retry Counter (SSRC) and Long Retry Counter (SLRC).

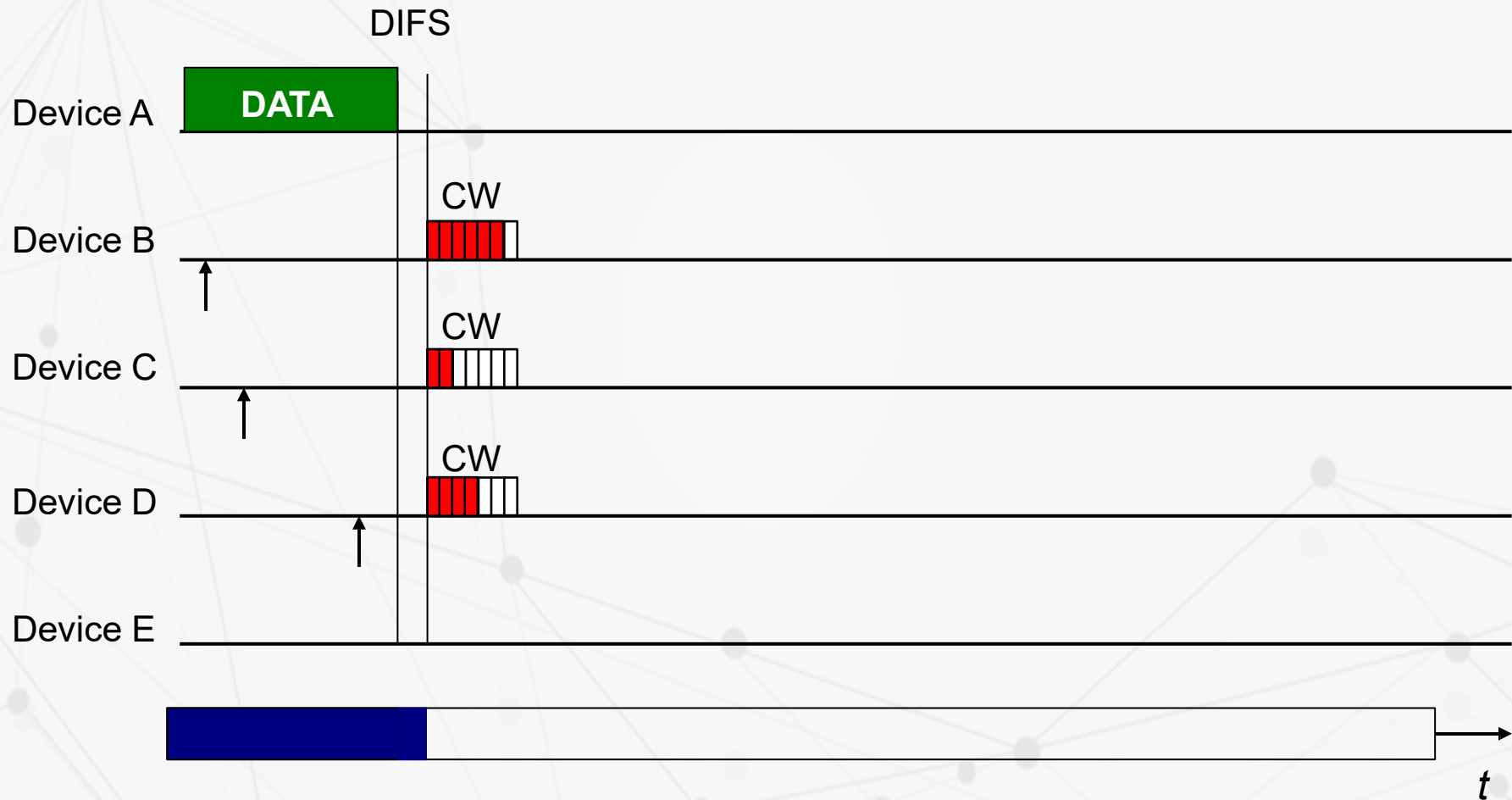
- SSRC increases by one for each failed RTS frame or data frame when RTS/CTS is not utilized for this transmission.
- SLRC increases by one for each failed data frame when RTS/CTS is utilized for this transmission.
- Both counter is reset upon a successful transmission.
- Retry is aborted when SSRC or SLRC reach the defined retry limit. The station then proceeds with the transmission of the next frame on the queue.

Question for discussion: Why IEEE 802.11 considers a retry limit for frame transmissions?



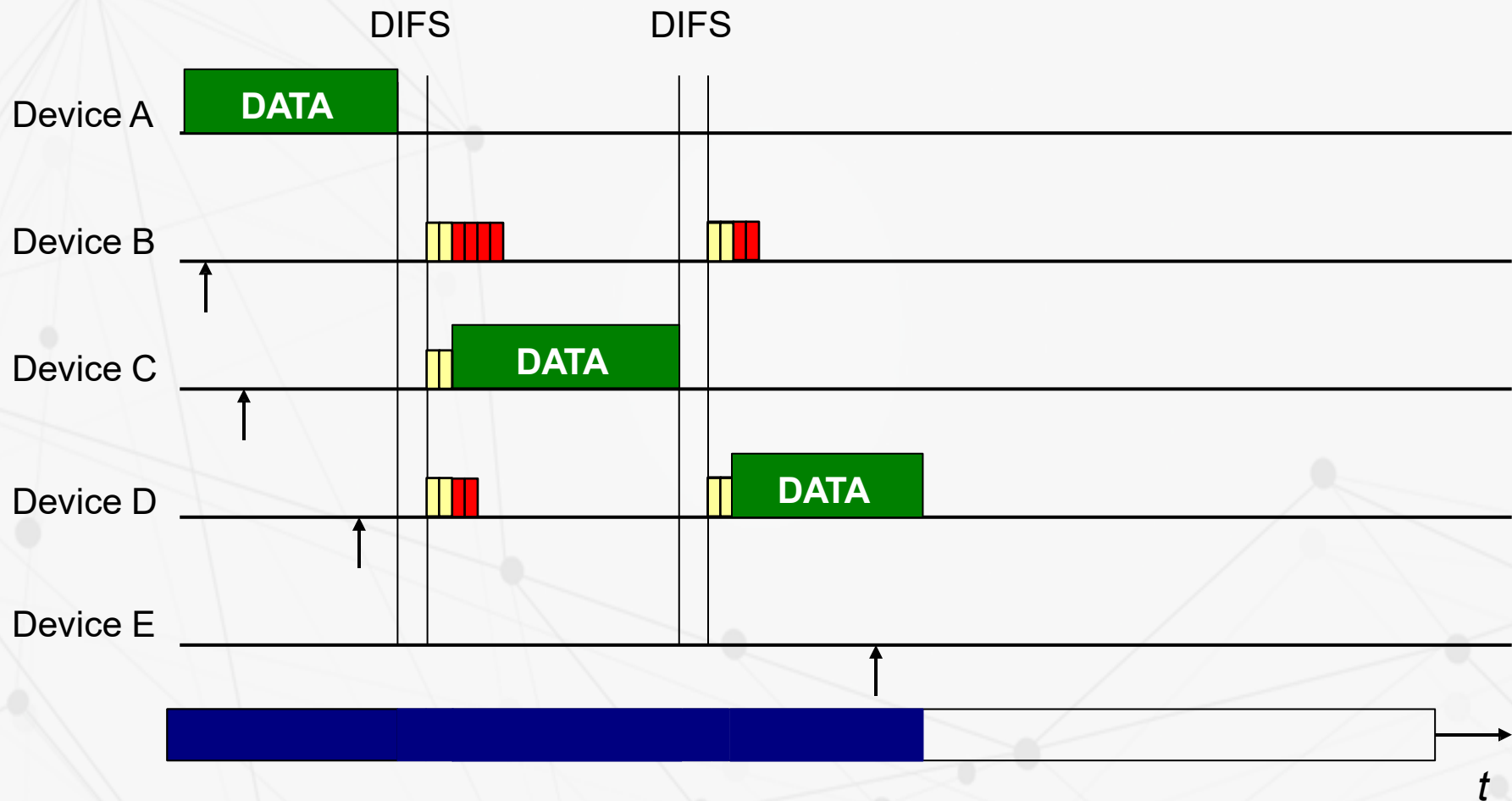
IEEE 802.11 MAC sub-layer

Example of DCF operation



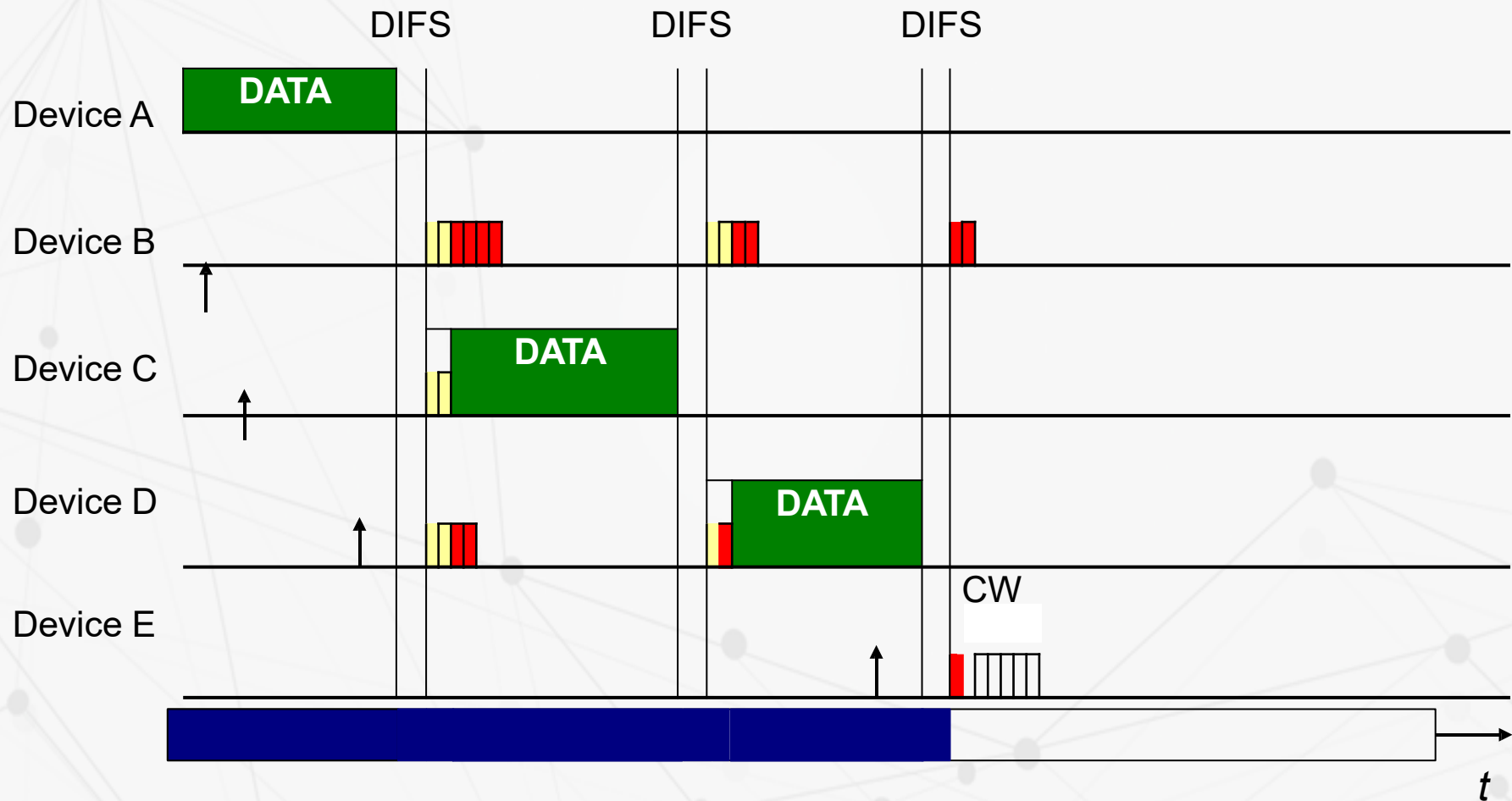
IEEE 802.11 MAC sub-layer

Example of DCF operation



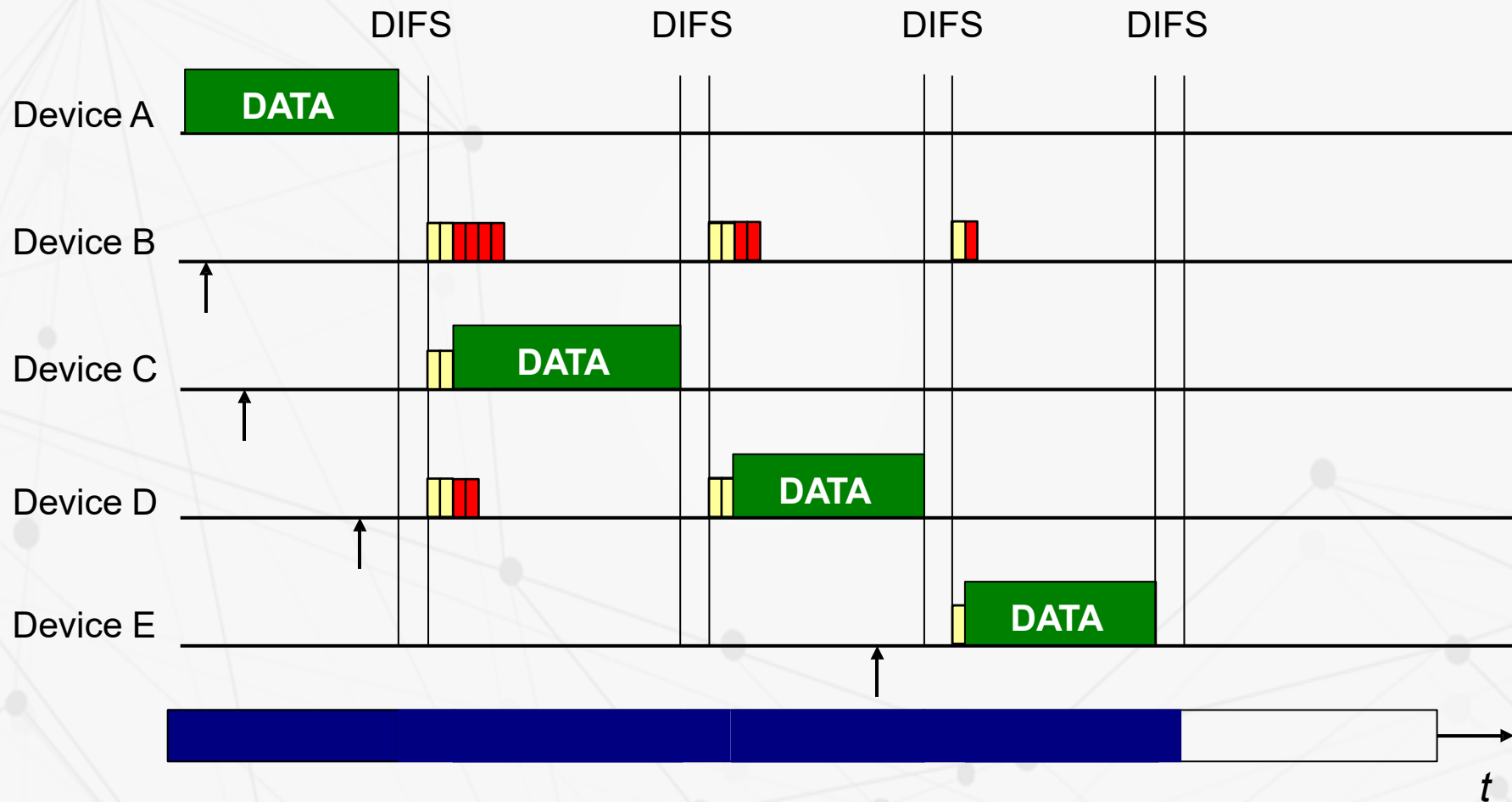
IEEE 802.11 MAC sub-layer

Example of DCF operation



IEEE 802.11 MAC sub-layer

Example of DCF operation



IEEE 802.11 MAC sub-layer

Example of DCF operation

